Introduction to Algebra and Analysis

Part **Foundations**

Notes originally by A. Blanco and I. Todorov

v 2020-10-04

Chapter 1

Number systems

We shall start our course with a review of some of the fundamental properties of the different number systems, ranging from the natural to the complex numbers. A good part of the material of this chapter is probably known to you from school.

1.1 Sets

Our starting point is to give some terminology. The basic item of modern mathematics is a **set**, which we think of as a collection of distinct objects.

Examples 1.1.1.

- The set of numbers 1,2 and 3. We write this as $\{1, 2, 3\}$.
- The set of students in this class.
- The set of student numbers.

We can of course have infinite sets.

Definition 1.1.2. *The set of **natural numbers** is the set* $\mathbb{N} = \{1, 2, 3, \dots\}$.
*The set of **integers** is the set* $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.

There are, in general, several ways of describing a set. However, there are three, which are the most common ones. They are:

– by writing down explicitly, between curly brackets, the list of all its elements. For instance, $\{\Box, \Diamond, 3, 11, \aleph\}$ stands for the set whose elements are precisely the symbols listed.

– by writing between curly brackets a formula to compute its elements (in this case, one also needs to specify the range of the parameters in the formula). For instance, $\{2k : k \in \mathbb{N}\}$ (to be read *"numbers of the form $2k$ with k in $\mathbb{N}$"*) represents the set of all even natural numbers.

– if our set happens to be a subset of a known set then, by indicating this together with the property that distinguishes its elements amongst the others between curly brackets. For instance, $\{k \in \mathbb{N} : 2 \big| k\}$ (to be read *"all natural numbers such that $2 \big| k$"*) also represents the set of all even natural numbers.

We also use $|$ in place of :, it is still read as 'such that'.

Facts about sets

- Sets do not come with an ordering, So $\{1,2,3\}$ is the same set as $\{3,1,2\}$.
- A set either contains an object or it does not. We do not think of a set as having repeats.
$$\{1,2,3,4,5,3\} = \{1,2,3,4,5\}$$
- It can be useful to write sets in such a way as they appear to have repeats:
$$\{1 + (-1)^n \mid n \in \mathbb{N}\} = \{0,2\}$$
- We can have sets with multiple indexes, for example
$$\left\{\cos\left(\frac{i\pi}{12}\right) + \sin\left(\frac{j\pi}{12}\right) \mid i,j \in \mathbb{N}\right\}.$$
It can be easier to work with this description of the set rather than just the list of its elements.

We shall refer to a subcollection of a given set as a **subset**. The objects of a set are called its **elements**. The notation $a \in A$ (or $A \ni a$) means that a is an element of the set A and is read as "*a belongs to A*". If a is not an element of A then we write $a \notin A$ (or $A \not\ni a$) and read it as "*a does not belong to A*".

Caution: The set $\{\{a,b\},c,d\}$ does not contain a. Its elements are $\{a,b\}$, c and d. Put simply, $a \neq \{a\}$.

Remark 1.1.3. With this simple interpretation of sets we can get into trouble.

Consider the set of all sets which do not contain themselves. This is an apparent paradox. The remedy is that 'set of all sets which do not contain themselves' is not a set. Similarly, the set of all sets is not a set.

The good news is that we won't actually encounter this problem in the course and as long as you stay away from constructions like the 'set of all sets', you won't run into any difficulty.

If you want, you can go read about the ZFC axioms for set theory, but that is not part of this course.

We'll return to set theory a little later on.

1.2 The natural numbers ($\mathbb{N}$)

We've already defined these above. There is no consensus about whether zero should be a natural number. For the purposes of this course, it won't be.

What can we do with the natural numbers? We have addition, multiplication and we can sometimes divide.

Definition 1.2.1. *A natural number a **divides a natural number** b (or that a **is a divisor of** b) if there exists a natural number c such that $b = ca$. If a and b are natural numbers such that a divides b we write this as $a \big| b$.*

Don't get confused between a divides b and a divided by b. One is a statement, the other is a number.

Don't get confused between $|$ in the definition of a set and $|$ in the statement of a divides b.

In a sense, made precise by the following result, the primes could be thought of as the building blocks of all natural numbers.

A natural number p is said to be a **prime number** (or just a **prime**) if $p \neq 1$ and its only divisors are 1 and p itself. Amusingly, there is no standard notation for the set of primes.

Theorem 1.2.2 (Fundamental Theorem of Arithmetic)**.** *Every natural number greater than 1 can be written as a product of prime numbers. Moreover, except for the order of the factors, this can be done in a unique way.*

So, for instance, $4 = 2 \cdot 2$, $6 = 2 \cdot 3$, $8 = 2 \cdot 2 \cdot 2$, $9 = 3 \cdot 3$, $10 = 2 \cdot 5$, $12 = 2 \cdot 2 \cdot 3 = 3 \cdot 2 \cdot 2$, and so on. Of course, the decomposition of a prime number contains only one factor – the prime number itself.

But, how do we know that the statement of Theorem 1.2.2 is true? A few words seem to be in order at this point.

There are two aspects to the solution of any mathematical problem. On the one hand, the answer to the problem, which may come in the form of a number, a formula, an example, etc., and, on the other hand, the explanation of how did you arrive at your answer. This explanation, whose aim is simply to convince another party that your answer is the right one, is what we call a mathematical proof (and for a mathematician, this is the most important part of the solution).

We can give proofs as we can agree our definitions. Since we have a common starting point, we should make sure that our arguments are unassailable: this is known as **mathematical rigour**. Chemists and physicists have to work hard to agree on anything, and eventually come to a consensus. Economists agree with whoever pays them (much like accountants) and politicians can't agree on anything.

Throughout this course, whenever possible (by which we mean, if the argument is short and simple) we shall present proofs for our results. The proof of Theorem 1.2.2, that we have in mind, uses a very important technique, closely related to the order structure of $\mathbb{N}$. We shall discuss this technique in the final part of this section, and then use it to prove Theorem 1.2.2.

Now, coming back to the prime numbers, our next proposition tells us another useful fact connected with them. We shall derive it from Theorem 1.2.2.

Proposition 1.2.3. *Let p, a and b be natural numbers. If p is a prime number and $p \big| ab$ then $p \big| a$ or $p \big| b$.*

Proof. If $a = 1$ or $b = 1$, then the claim clearly holds, thus we assume further that $a \neq 1$ and $b \neq 1$. Since $p \big| ab$ there is $c \in \mathbb{N}$ such that $ab = cp$ $(*)$; notice that c cannot be 1. By Theorem 1.2.2, there are prime numbers $p_1, p_2, \ldots, p_k, q_1, q_2, \ldots, q_l, r_1, r_2, \ldots, r_m$ such that $a = p_1 p_2 \ldots p_k$, $b = q_1 q_2 \ldots q_l$ and $c = r_1 r_2 \ldots r_m$. Substituting a, b and c in $(*)$ by their representations we find that $p_1 \ldots p_k q_1 \ldots q_l = r_1 \ldots r_m p$. By Theorem 1.2.2 (again), the factors on both sides of this last identity must be the same. So, $p = p_i$ for some $1 \le i \le k$, in which case $p \big| a$, or $p = q_i$ for some $1 \le i \le l$, in which case, $p \big| b$. □

Fairly often, mathematical statements appear or can be written in the form "A implies B", where A and B are also statements, which can be (independently) true or false. This can be re-written in the form if A is true, then B is true or whenever A is true, so is B. We'll learn this kind of language by using it lectures and practicing it on the homeworks we have plenty of time.

For instance, $\big(n, m \in \mathbb{N}$ are such that $n - m \ge 2\big)$ implies $\big(n^2 - m^2$ is not a prime$\big)$.

Similarly, if Nicholas Cage is in a film, then that film will be terrible.

Now, there are essentially three approaches to verifying the correctness of a mathematical statement of the form "A implies B". They are:

> *Directly:* One tries to deduce B from A, by arguing in a logical way. The proof of Proposition 1.2.3 above is of this kind.
>
> *Indirectly:* One tries to show that if B fails to be true then A must fail to be true as well (of course, arguing in a logical way). Sometimes also called the **contrapositive**.
>
> *By contradiction:* One assumes A is true but B is not, and tries to achieve a contradiction (again, arguing in a logical way).

Which approach should you use is something that will come to you with practice.

Now, this is how Euclid's proof (by contradiction!) of the infinitude of primes goes. The advantage of contradiction is that we have two facts! That A is true and that B is not true.

Proposition 1.2.4. *There are infinitely many primes.*

Proof. Suppose towards a contradiction that there are only finitely many prime numbers, $p_1, p_2, \ldots, p_k$ say. Then consider the number

$$p = p_1 \cdot p_2 \cdot \ldots \cdot p_k + 1.$$

The number p is a natural number which is not divisible by any of the prime numbers $p_1, p_2, \ldots, p_k$. But, by the Fundamental Theorem of Arithmetic, p has to be divisible by at least one of $p_1, \ldots, p_k$, a contradiction. □

The contrapositive (indirect proof) is useful to rewrite a proof into something that can be simpler to show.

Lemma 1.2.5. *If x is irrational then $-x$ is irrational.*

Proof. We use an indirect proof and show that "If $-x$ is NOT irrational then x is NOT irrational".

But that statement is exactly: "If $-x$ is rational then x is rational", which is clearly true. □

This is also easy to approach by contradiction: if x is irrational and $-x$ is rational, then $x = -(-x)$ is rational and irrational, which is a contradiction.

It seems convenient at this point to introduce some more terminology. You may find it a bit confusing at the beginning but you will get used to it! In mathematics, a statement of the form *"A implies B"* is commonly written as "A $\Rightarrow$ B". The symbol $\Rightarrow$ means that B is a consequence of A. The expression "A $\Rightarrow$ B" is read as *"A implies B"* (or *"A is sufficient for B"*, or *"A only if B"*). A similar meaning is attached to "A $\Leftarrow$ B", which is read as *"B implies A"* (or *"A is necessary for B"*, or *"A if B"*). If A $\Rightarrow$ B and A $\Leftarrow$ B then we write A $\iff$ B, and read it as *"A is equivalent to B"* (or *"A is necessary and sufficient for B"*, or *"A if and only if B"*).

You should try to avoid overusing symbols like $\Rightarrow$, it is easier to read something written in full sentences rather than a pile of symbols.

1.2.1 The method of mathematical induction

In this subsection, we discuss a method of direct mathematical proof, intimately connected with the order structure of $\mathbb{N}$. It is known as **Mathematical Induction**. It is (usually) useful if one is trying to show that a particular statement, that depends on a variable n taking its values in $\mathbb{N}$, holds true for every n greater than or equal to a certain m.

How can we prove such a thing? We could write an infinitely long proof, one for each $n > m$.

We could just check a bunch of values, and convince ourself it is a pattern. But this isn't rigourous.

- $2^{2^1} + 1 = 5$, $2^{2^2} + 1 = 17$, $2^{2^3} + 1 = 257$ and $2^{2^4} + 1 = 65537$ are all prime numbers. This led Fermat, in the 17th century, to believe that every number of this form was a prime. However, a hundred years later, Euler proved Fermat to be wrong by showing that $2^{2^5} + 1 = 4294967297 = 641 \cdot 6700417$.
- $3 \big| (n^3 - n)$ for every $n \in \mathbb{N}$; $5 \big| (n^5 - n)$ for every $n \in \mathbb{N}$; and $7 \big| (n^7 - n)$ for every $n \in \mathbb{N}$. This led Leibniz, in the 17th century, to conjecture that $k \big| (n^k - n)$ for every $n \in \mathbb{N}$, whenever k is an odd number. However, as he discovered later on, the pattern breaks down for 9 because $2^9 - 2 = 510$, which is not divisible by 9.
- Here is a conjecture: for $k > 1$, if $x_1^n + \cdots + x_k^n = z^n$ has a solution in the positive integers then $n \leqslant k$. The smallest counterexample for $n = 4$ is

$$31858749840007945920321 = 95800^4 + 217519^4 + 414560^4 = 422481^4$$

The method can be formally stated as follows:

> *First, show the result holds for m.*
>
> *Second, assume the result holds for some n greater than or equal to m, and show that it holds for $n + 1$.*

If one is able to carry out both steps then the statement must be true for every natural number $\geq m$.

Example 1.2.6. Show that $1+3+5+\cdots+(2n-1)=n^2$ for every $n\in\mathbb{N}$.

First: $1=1^2$, so the identity holds for $n=1$.

Second: Suppose the identity holds for some $n\geq 1$. Then

$$1+3+5+\cdots+(2n-1)+(2n+1)=n^2+(2n+1)=(n+1)^2,$$

so the identity holds for $n+1$.

By mathematical induction, the result is true for every n.

Example 1.2.7. Show that $1^3+2^3+3^3+\cdots+n^3=\left[\dfrac{n(n+1)}{2}\right]^2$ for every $n\in\mathbb{N}$.

First: $1^3=\left[\dfrac{1(1+1)}{2}\right]^2$, so the identity holds for $n=1$.

Second: Suppose the identity holds for some $n\geq 1$. Then

$$\begin{aligned}1^3+2^3+\cdots+n^3+(n+1)^3 &= \left[\frac{n(n+1)}{2}\right]^2+(n+1)^3 = (n+1)^2\left(\frac{n^2}{4}+(n+1)\right)\\ &= \frac{(n+1)^2(n^2+4n+4)}{4} = \frac{(n+1)^2(n+2)^2}{4}\\ &= \left[\frac{(n+1)((n+1)+1)}{2}\right]^2,\end{aligned}$$

so the identity holds for $n+1$.

By mathematical induction, the result is true for every n.

Example 1.2.8. Show that

$$\frac{1}{1\cdot 5}+\frac{1}{5\cdot 9}+\frac{1}{9\cdot 13}+\cdots+\frac{1}{(4n-3)(4n+1)}=\frac{n}{4n+1}$$

for every $n\in\mathbb{N}$.

First: $\dfrac{1}{1\cdot 5}=\dfrac{1}{4\cdot 1+1}$, so the identity holds for $n=1$.

Second: Suppose the identity holds for some $n\geq 1$. Then

$$\begin{aligned}&\frac{1}{1\cdot 5}+\frac{1}{5\cdot 9}+\cdots+\frac{1}{(4n-3)(4n+1)}+\frac{1}{(4(n+1)-3)(4(n+1)+1)}\\ &\qquad = \frac{n}{4n+1}+\frac{1}{(4n+1)(4n+5)} = \frac{1}{(4n+1)}\left(n+\frac{1}{4n+5}\right)\\ &\qquad = \frac{4n^2+5n+1}{(4n+1)(4n+5)} = \frac{(4n+1)(n+1)}{(4n+1)(4n+5)} = \frac{n+1}{4(n+1)+1},\end{aligned}$$

so the identity holds for $n+1$.

By mathematical induction, the result is true for every n.

Example 1.2.9. Show that

$$1\cdot 2\cdot 3+2\cdot 3\cdot 4+3\cdot 4\cdot 5+\cdots+n(n+1)(n+2)=\frac{n(n+1)(n+2)(n+3)}{4}$$

for every $n \in \mathbb{N}$.

First: $1 \cdot 2 \cdot 3 = \dfrac{1(1+1)(1+2)(1+3)}{4}$, so the identity holds for $n = 1$.

Second: Suppose the identity holds for some $n \geq 1$. Then

$$\begin{aligned}
&1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \cdots + k(k+1)(k+2) + (k+1)((k+1)+1)((k+1)+2) \\
&\qquad = \frac{k(k+1)(k+2)(k+3)}{4} + (k+1)(k+2)(k+3) \\
&\qquad = (k+1)(k+2)(k+3)\left(\frac{k}{4}+1\right) = \frac{(k+1)(k+2)(k+3)(k+4)}{4} \\
&\qquad = \frac{(k+1)((k+1)+1)((k+1)+2)((k+1)+3)}{4},
\end{aligned}$$

so the identity holds for $n + 1$.

By mathematical induction, the result is true for every n.

Example 1.2.10. Show that $\dfrac{1}{n+1} + \dfrac{1}{n+2} + \cdots + \dfrac{1}{2n} > \dfrac{13}{24}$ for every $n \in \mathbb{N}$ greater than or equal to 2.

First: $\dfrac{1}{2+1} + \dfrac{1}{2+2} = \dfrac{1}{3} + \dfrac{1}{4} = \dfrac{7}{12} = \dfrac{14}{24} > \dfrac{13}{24}$, so the inequality holds for $n = 2$.

Second: Suppose the inequality holds for some $n \geq 2$. Then

$$\begin{aligned}
&\frac{1}{(n+1)+1} + \frac{1}{(n+1)+2} + \cdots + \frac{1}{2(n+1)} \\
&\qquad = \frac{1}{n+2} + \frac{1}{n+3} + \cdots + \frac{1}{2n} + \frac{1}{2n+1} + \frac{1}{2n+2} \\
&\qquad = \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}\right) - \frac{1}{n+1} + \frac{1}{2n+1} + \frac{1}{2n+2} \\
&\qquad > \frac{13}{24} + \left(\frac{1}{2n+1} + \frac{1}{2n+2} - \frac{1}{n+1}\right) \\
&\qquad > \frac{13}{24} + \left(\frac{1}{2n+2} + \frac{1}{2n+2} - \frac{1}{n+1}\right) = \frac{13}{24},
\end{aligned}$$

so the inequality holds for $n + 1$.

By mathematical induction, the result is true for every $n \geq 2$.

Example 1.2.11. Show that $7^{2n-1} + 13^{n+1} - 8$ is divisible by 12 for every $n \in \mathbb{N}$.

First: $7 + 13^2 - 8 = 13^2 - 1 = (13-1)(13+1) = 12 \cdot 14$, so the claim is true for $n = 1$.

Second: Suppose the claim holds for some $n \geq 1$. Then

$$\begin{aligned}
7^{2n+1} + 13^{n+2} - 8 &= 49(7^{2n-1}) + 13(13^{n+1}) - 8 \\
&= 48(7^{2n-1}) + 12(13^{n+1}) + (7^{2n-1} + 13^{n+1} - 8) \\
&= 12(4(7^{2n-1}) + 13^{n+1}) + (7^{2n-1} + 13^{n+1} - 8)
\end{aligned}$$

or

$$\begin{aligned}
7^{2n+1}+13^{n+2}-8 &= (7^{2n-1}+13^{n+1}-8)+(7^{2n+1}+13^{n+2}-8-7^{2n-1}-13^{n+1}+8)\\
&= (7^{2n-1}+13^{n+1}-8)+(7^{2n-1}(7^2-1)+13^{n+1}(13-1))\\
&= (7^{2n-1}+13^{n+1}-8)+(7^{2n-1}\cdot 48+13^{n+1}\cdot 12).
\end{aligned}$$

Both expressions in parentheses are divisible by 12, so the claim is also true for $n+1$.

By mathematical induction, the result is true for every n.

In some situations, it may be necessary to use **complete induction**. In this assume that the result is true, not just for some n greater than or equal to m, but for every natural number between m and n, in order to show that the result holds for $n+1$. This form of the method of Mathematical Induction is known as **Complete Induction**. Our last example illustrates the idea.

At last we can give the postponed proof of the Fundamental Theorem of Arithmetic. We will only show that every natural number greater than 2 has a decomposition, as claimed in the theorem, and omit the proof of the uniqueness statement, which is a bit long.

Proof. First: The result is true for 2, because 2 is a prime number.

Second: Suppose the result is true for every natural number greater than 1 and $\leq n$. We then need to show that the result is true for $n+1$. If $n+1$ is prime then we are finished. Otherwise, $n+1$ has to be divisible by some number different from 1 and itself, m say. So, $n+1=m\cdot k$, where k is also bigger than 1 and smaller than $n+1$. Since both m and k are greater than 1 and $\leq n$ they can both be written as products of primes. Say $m=p_1p_2\dots p_i$ and $k=q_1q_2\dots q_j$, where $p_1,\dots,p_i,q_1,\dots,q_j$ are all prime numbers. Then $n+1=p_1\dots p_iq_1\dots q_j$, as needed.

By complete induction, the result is true for every $n\geq 2$. □

1.3 The integers ($\mathbb{Z}$)

The fact that, in general, the difference of two natural numbers need not be a natural number means that an equation like $x+3=1$ could not be solved if the the only numbers available were the naturals. This leads us 'naturally' to the integers. The **integers** are the numbers …, -3, -2, -1, 0, 1, 2, 3, … Of course, there are more practical reasons for introducing negative numbers. Think, for example, of having to describe quantities that can naturally change (in principle, indefinitely) in two 'opposite directions', e.g., time – past and future.

The set of all integers is denoted by $\mathbb{Z}$. The sum, product and difference of two integers are again integers. It is worth pointing out that, although division of two integers is not always possible within $\mathbb{Z}$, it is always true that, given any two integers a and b, there are integers k and r, with $0\leq r<|a|$, such that $b=ka+r$.

1.4 The rational numbers ($\mathbb{Q}$)

We already mentioned that the result of dividing one integer by another need not be an integer. However, fairly often in real life one needs to consider parts of a whole. This leads one naturally to the concept of a rational number (or fraction).

A number x is called **rational** if it can be written as the quotient of two integers, i.e., in the form p/q, where p and q belong to $\mathbb{Z}$ and $q \neq 0$. The numbers p and q figuring in this representation are called the numerator and denominator, respectively, of the representation. We shall write $\mathbb{Q}$ for the set of all rational numbers.

Note that every integer is a rational number, for if p belongs to $\mathbb{Z}$ then p can be written as $p/1$. Also note that the same rational number can be represented in more than one way as the quotient of two integers, for if a, b, c and d are integers such that $ad = bc$, and b and d are different from zero, then $a/b = c/d$. So, for instance, $2/3 = 4/6 = 8/12 = \ldots$ and so on. A representation p/q is said to be **irreducible** if 1 and -1 are the only common factors of p and q. It is not hard to see that every rational number admits irreducible representations.

Rational numbers are added and multiplied according to the rules:

$$\frac{a}{b} + \frac{c}{d} := \frac{ad+bc}{bd} \qquad \text{and} \qquad \frac{a}{b} \cdot \frac{c}{d} := \frac{ac}{bd}.$$

It is clear from the last definition that every rational number p/q different from zero has a rational inverse[1], namely, the number q/p. One can then define the quotient of two rational numbers p/q and m/n (of course, provided $m/n \neq 0$) as

$$\frac{p}{q} \div \frac{m}{n} := \frac{p}{q} \cdot \frac{n}{m}.$$

Note that the rational numbers allow us to solve any equation of the form $m \cdot x = n$, in which m and n are integers and $m \neq 0$.

1.5 The real numbers ($\mathbb{R}$)

Imagine that you want to know the exact distance between two given points in space. It was already known to the ancient Greeks, that the rational numbers are not enough for this purpose. Precisely, not every length is a rational number. For instance, in an isosceles right–angled triangle whose legs are of length 1, the length of the hypotenuse is $\sqrt{1^2+1^2} = \sqrt{2}$ (by Pythagoras). However, $\sqrt{2}$ is not a rational number. To see this, one can argue by contradiction, as follows:

Lemma 1.5.1. *The number $\sqrt{2}$ is irrational.*

Proof. Suppose $\sqrt{2}$ is rational, i.e., there are integers p and q ($q \neq 0$) such that

$$\sqrt{2} = \frac{p}{q}. \tag{1.1}$$

[1] The inverse of a number r different from zero is the only number, usually denoted by r^{-1}, with the property that $r \cdot r^{-1} = 1$.

Clearly, we can assume p/q to be irreducible and both p and q to be positive. It follows from (1.1) that

$$2q^2 = p^2, \tag{1.2}$$

so $2 \big| p^2$, and by Proposition 1.2.3, $2 \big| p$. Thus, $p = 2r$ for some natural number r. Substituting p by $2r$ in (1.2) and simplifying one obtains that $q^2 = 2r^2$, and so, $2 \big| q$. But this means that 2 divides both p and q, contradicting the fact that p/q was chosen to be irreducible. □

So, if one wants to be able to measure the distance between any two points in space, other numbers apart from the rationals are needed. This larger collection are the **real numbers**. We shall denote the set of all real numbers by $\mathbb{R}$. Real numbers which are not rational are called **irrational**. So, for instance, $\sqrt{2}$ is an irrational number.

Informally, one could think of the real numbers as the decimal numbers or the points of a straight line. Precisely, in a horizontal line, choose a point, call it the origin, and then identify any other point of the line with the value of the distance between that point and the origin (distances are taken with negative sign for points to the left of the origin and with positive sign for points to the right). This geometric representation of the real numbers is commonly known as **the real line**. Although the geometric interpretation of the real numbers as the points of a straight line is particularly appealing, to do mathematics, one needs a more workable definition of a real number.

Here are two NON-EXAMINABLE definitions.

Definition 1.5.2. *The real numbers are the unique (up to isomorphism) FIELD $\mathbb{R}$ which is ORDERED:*

- *if $x \geq y$ then $x + z \geq y + z$*
- *if $x \geq 0$ and $y \geq 0$ then $xy \geqslant 0$*

and every non-empty subset which is bounded above in $\mathbb{R}$ has a LEAST UPPER BOUND IN $\mathbb{R}$.

Definition 1.5.3. *The real numbers are the set of EQUIVALENCE CLASSES of CAUCHY sequences of rational numbers under the relation $(a_n)_{n\in\mathbb{N}} \simeq (b_n)_{n\in\mathbb{N}}$ if $(a_n - b_n)_{n\in\mathbb{N}}$ converges to zero.*

So for the time being, here is a working definition.

> *The real numbers are a collection of numbers, containing the rational numbers, obeying the same arithmetic and order rules as the rational numbers, and with a property, not shared by the rationals, known as the Completeness Axiom.*

To explain the Completeness Axiom, we first need to introduce a few concepts.

Definition 1.5.4. *We will say that a subset A of $\mathbb{R}$ is*

i) ***bounded above*** *if there is a real number u such that $a \leq u$ for every $a \in A$. Such number u will be called an **upper bound** for A.*

ii) ***bounded below*** *if there is a real number l such that $l \leq a$ for every $a \in A$. Such number l will be called a* ***lower bound*** *for A.*

iii) ***bounded*** *if it is bounded above and below.*

Example 1.5.5.

- The set $\mathbb{Z}$ is neither bounded below nor bounded above.
- The set $\mathbb{N}$ is not bounded above, however, it is bounded below. Indeed, 1 is clearly a lower bound for $\mathbb{N}$ because $1 \leq n$ for every $n \in \mathbb{N}$.
- Every finite subset of $\mathbb{R}$ is clearly bounded. Take as upper bound its greatest element and as lower bound its smallest element.
- The set $\{x \in \mathbb{R} : x^2 \leq 1\}$ is bounded, because $x^2 \leq 1 \iff -1 \leq x \leq 1$. So 1 is an upper bound for the set and -1 is a lower bound. But $3\pi \times 10^6$ is also an upper bound. Is this a sensible upper bound?
- The set $\{m/n : m, n \in \mathbb{N} \text{ and } m+n \leq 10\}$ is finite because $(m+n \leq 10) \Rightarrow (m \leq 10$ and $n \leq 10)$. Therefore, it must be bounded.

We want to codify the idea of the best upper bound. The upper bound which is as small as it can be and still be an upper bound.

Definition 1.5.6 (Supremum)**.** *Let A be a subset of $\mathbb{R}$ with at least one element and bounded above. A real number $\mathfrak{s}$ is said to be the* ***supremum*** *of A if*

i) *$\mathfrak{s}$ is an upper bound for A; and*

ii) *For every $\varepsilon > 0$ there is an a in A such that $\mathfrak{s} - \varepsilon < a$.*

The number $\mathfrak{s}$ shall be denoted by $\sup A$.

This is the toughest definition in this part of the course. We need to understand this definition there are several bits we want to do:

- how to understand and visualise this concept
- how to use the abstract language.

The idea is that it is the "smallest upper bound", of all upper bounds, this is then smallest. The definition states that is an upper bound, the $\mathfrak{s} - \varepsilon$ part is telling us that it is the smallest. The point is that the number $\mathfrak{s} - \varepsilon$ is a way of writing something smaller than $\mathfrak{s}$. We are then saying that $\mathfrak{s} - \varepsilon$ is NOT an upper bound as there is some element of S which is bigger than $\mathfrak{s} - \varepsilon$.

Let's go through some examples, getting harder as we go.

Example 1.5.7. Let $S := \{1, 2, 3\}$. Then $\sup S = 3$. Indeed,

(i) $x \leq 3$ for every $x \in S$, so 3 is an upper bound.

(ii) Let $\varepsilon > 0$ arbitrary (THIS MEANS THAT WHAT FOLLOWS SHOULD WORK FOR ANY ε) then $3 - \varepsilon < 3$ and $3 \in S$, so 3 is the supremum.

Now let's go through one that is a little harder.

Example 1.5.8. Let a be a real number and let $S := \{x \in \mathbb{R} : x < a\}$. Then $\sup S = a$. Indeed,

(i) $x \leq a$ for every $x \in S$, so a is an upper bound.We take the convention that if you are less than a, then you are less than or equal to a, so $x < a$ implies $x <\leqslant a$.

Draw the real line, $a - \varepsilon$, and c.

(ii) Let $\varepsilon > 0$ arbitrary and let $c = a - \frac{1}{2}\varepsilon$. Then the real number $c < a$ is in S and $a - \varepsilon < a - \frac{1}{2}\varepsilon = c$ so a is the supremum.

Let's do a similar example that is a bit harder.

Example 1.5.9. Let a and b be real numbers, $a < b$, and let $S := \{x \in \mathbb{R} : a < x < b\}$. Then $\sup S = b$. Indeed,

(i) $x \leq b$ for every $x \in S$, so b is an upper bound.

Draw the real line, talk about we want to look at $b - \varepsilon$, split into two cases

(ii) Let $\varepsilon > 0$ arbitrary and then either $b - \varepsilon \leqslant a$ or $a < b - \varepsilon$.

In the first case consider $(a + b)/2$, then

$$b - \varepsilon \leqslant a = (a + a)/2 < (a + b)/2 < (b + b)/2 = b$$

so $(a + b)/2 \in S$ and it is bigger than $b - \varepsilon$.

In the second case, consider $b - \varepsilon/2$, then

$$a < b - \varepsilon < b - \varepsilon/2 < b$$

so $b - \varepsilon/2 \in S$ and it is bigger than $b - \varepsilon$.

We could do this in one step: let $c = max\{a, b - \varepsilon\}$. Then the number $r := (b + c)/2$ satisfies $b - \varepsilon < r$ and also $a < r < b$, so it belongs to S. But this is harder to see straight away.

Example 1.5.10. Let $S := \{1/n : n \in \mathbb{N}\}$.

Draw on real line and also as a the graph of a sequence.

Then $\sup S = 1$. Indeed,

(i) $1/n \leq 1$ for every $n \in \mathbb{N}$, so 1 is an upper bound for S.

(ii) Let $\varepsilon > 0$ arbitrary. Then $1 - \varepsilon < 1$ and $1 \in S$, so we are done.

It can be hard to work out what the supremum should be. Drawing a picture can help, as can calculating some elements of the set. Later we will give some rules that will make this much easier.

Example 1.5.11. Let $S := \{-1/n : n \in \mathbb{N}\}$.

Draw on real line and also as a the graph of a sequence.

Then $\sup S = 0$. Indeed,

(i) $-1/n < 0$ for every $n \in \mathbb{N}$, so 0 is an upper bound for S.

(ii) Let $\varepsilon > 0$ arbitrary and choose $n \geq 1/\varepsilon$. Then $0 - \varepsilon = -\varepsilon < -1/n$ and $-1/n \in S$.

The big question here is how did I find that condition on ε? I'll explain in the next example.

Example 1.5.12. Let $S := \{18 - 3/n : n \in \mathbb{N}\}$. Then $\sup S = 18$. Indeed,

(i) $18 - 3/n < 18$ for every $n \in \mathbb{N}$, so 18 is an upper bound.

(ii) Let $\varepsilon > 0$ arbitrary and choose $n > \boxed{}$. Then $18 - \varepsilon < 18 - 3/n$ and $18 - 3/n \in S$.

To see this,

we know $n > 3/\varepsilon$
so $\varepsilon > 3/n$
hence $-\varepsilon < -3/n$
thus $18 - \varepsilon < 18 - 3/n$

> We want
>
> $$18 - \varepsilon < 18 - 3/n \Leftrightarrow$$
> $$-\varepsilon < -3/n \Leftrightarrow$$
> $$\varepsilon > 3/n \Leftrightarrow$$
> $$n\varepsilon > 3 \Leftrightarrow$$
> $$n > 3/\varepsilon$$
>
> Now we put $n > 3/\varepsilon$ into the box!

It is OK to be confused, this is hard and needs you to practice. So come to the tutorials and the maths mentoring session.

Example 1.5.13. Let $S := \left\{ \dfrac{(-1)^n n}{2n+1} : n \in \mathbb{N} \right\}$. Then $\sup S = 1/2$. Indeed,

(i) $(-1)^n n/(2n+1) \leq n/(2n) \leq 1/2$, so $1/2$ is an upper bound for S.

(ii) Let $\varepsilon > 0$ arbitrary and choose $n > \boxed{}$. Then $1/2 - \varepsilon < 1/2 - 1/(2(2n+1)) = n/(2n+1) = (-1)^n n/(2n+1)$ and $(-1)^n n/(2n+1) \in S$. To see this,

To see this, As n is even, we know

we know $1/(4\varepsilon) - 1/2 < n$
so $1/(2\varepsilon) < 2n + 1$
and $1/2 < (2n+1)\varepsilon$
thus $n + 1/2 - (2n+1)\varepsilon < n$
so $1/2 - \varepsilon < n/(2n+1)$

> We may as well make n even. We want
>
> $$1/2 - \varepsilon < n/(2n+1) \Leftrightarrow$$
> $$n + 1/2 - (2n+1)\varepsilon < n \Leftrightarrow$$
> $$1/2 - (2n+1)\varepsilon < 0 \Leftrightarrow$$
> $$1/2 < (2n+1)\varepsilon \Leftrightarrow$$
> $$1/(2\varepsilon) < 2n+1 \Leftrightarrow$$
> $$1/(4\varepsilon) - 1/2 < n$$
>
> Now we put $n > 1/(4\varepsilon) - 1/2$ and n even into the box!

Now that we understand suprema, we can give the important fact about the real numbers:

The **Completeness Axiom** states:

> *Every bounded above subset of $\mathbb{R}$ with at least one element has a supremum.*

Note that a supremum is always a real number.

The next theorem contains some very useful facts about the supremum.

Theorem 1.5.14. *Let A and B be subsets of $\mathbb{R}$ containing at least one element and bounded above.*

i) *If* $S := \{a + b : a \in A \text{ and } b \in B\}$ *then* $\sup S = \sup A + \sup B$.

ii) *If A and B only contain positive real numbers and* $P := \{ab : a \in A \text{ and } b \in B\}$ *then* $\sup P = \sup A \cdot \sup B$.

iii) *For every real number $c > 0$, if $C := \{ca : a \in A\}$ then* $\sup C = c \sup A$.

Proof. Here, we only prove (i). The proofs of (ii) and (iii) will be left as Tutorial exercises.

i) Let $\alpha := \sup A$ and $\beta := \sup B$. We show that $\alpha + \beta$ verifies both conditions of Definition 1.5.6 with respect to the set S.

First, by our definitions of α and β, $a \leq \alpha$ for every $a \in A$ and $b \leq \beta$ for every $b \in B$. Combining both inequalities, we see that $a + b \leq \alpha + \beta$ for every $a \in A$ and every $b \in B$, so the first condition of Definition 1.5.6 holds.

Second, let $\varepsilon > 0$ arbitrary. Again, by our definitions of α and β, there exist $a \in A$ and $b \in B$ such that $\alpha - \varepsilon/2 < a$ and $\beta - \varepsilon/2 < b$. Adding the two inequalities, we obtain that $(\alpha + \beta) - \varepsilon < a + b$, and since $a + b \in S$, the second condition holds too. □

Example 1.5.15. Let $S = \{\frac{2+3m}{mn} : m, n \in \mathbb{N}\}$. Then

$$\begin{aligned}
\sup S &= \sup\left\{\frac{1}{n}\left(\frac{2+3m}{m}\right) : m, n \in \mathbb{N}\right\} \\
&= \sup\{1/n : n \in \mathbb{N}\} \cdot \sup\big\{(2+3m)/m : m \in \mathbb{N}\big\} \\
&= \sup\{1/n : n \in \mathbb{N}\} \cdot \sup\{2/m + 3 : m \in \mathbb{N}\} \\
&= \sup\{1/n : n \in \mathbb{N}\} \cdot \big(2 \cdot \sup\{1/m : m \in \mathbb{N}\} + \sup\{3\}\big) \\
&= 1 \cdot (2 \cdot 1 + 3) = 5.
\end{aligned}$$

There is a notion dual to that of supremum. It is called infimum.

Definition 1.5.16 (Infimum)**.** *Let A be a subset of $\mathbb{R}$ with at least one element and bounded below. A real number $\mathfrak{i}$ is said to be the* ***infimum*** *of A if*

i) *$\mathfrak{i}$ is a lower bound for A; and*

ii) *For every $\varepsilon > 0$ there is an a in A such that $a < \mathfrak{i} + \varepsilon$.*

The number $\mathfrak{i}$ shall be denoted by $\inf A$.

Example 1.5.17. Let a and b be real numbers, $a < b$, and let $S := \{x \in \mathbb{R} : a < x < b\}$. Then $\inf S = a$. Indeed,

(i) $a \leq x$ for every $x \in S$, so a is a lower bound.

(ii) Let $\varepsilon > 0$ arbitrary and let $c = \min\{a + \varepsilon, b\}$. Then $r := (a + c)/2$ satisfies $r < a + \varepsilon$ and also $a < r < b$, so it belongs to S.

Example 1.5.18. Let $S = \{1/n : n \in \mathbb{N}\}$. Then $\inf S = 0$. Indeed,

(i) $0 \leq 1/n$ for every $n \in \mathbb{N}$, so 0 is a lower bound for S.

(ii) Let $\varepsilon > 0$ arbitrary and choose $n > 1/\varepsilon$. Then $0 + \varepsilon = \varepsilon > 1/n$ and $1/n \in S$.

Example 1.5.19. Let $S = \{18 - 3/n : n \in \mathbb{N}\}$. Then $\inf S = 18 - 3 = 15$. Indeed,

(i) $18 - 3/n \geq 18 - 3 = 15$ for every $n \in \mathbb{N}$, so 15 is a lower bound for S.

(ii) Let $\varepsilon > 0$ arbitrary. Then $15 + \varepsilon > 15$ and $15 = 18 - 3/1 \in S$.

Supremum and infimum are related as follows.

Proposition 1.5.20. *Let A be a subset of $\mathbb{R}$ with at least one element and bounded below. Then $\inf A = -\sup(-A)$, where $-A := \{-a : a \in A\}$.*

Proof. Let $\alpha := \inf A$. We show that $-\alpha = \sup(-A)$. For this, we show that α satisfies both conditions of Definition 1.5.6.

First, by our definition of α, $\alpha \leq a$ for every $a \in A$, so $-a \leq -\alpha$ for every $a \in A$. Thus, the first condition holds.

Second, let $\varepsilon > 0$ arbitrary. Again, by our definition of α, there exists $a \in A$ such that $\alpha + \varepsilon > a$. Multiplying this inequality by -1, we obtain that $(-\alpha) - \varepsilon < -a$. Since $-a \in -A$, the second condition holds and we are done. $\square$

Example 1.5.21. Let $S := \{18 - 3/n : n \in \mathbb{N}\}$. Then $\inf S = -\sup(-S) = -\sup\{-18 + 3/n : n \in \mathbb{N}\} = -\big(\sup\{-18\} + 3 \cdot \sup\{1/n : n \in \mathbb{N}\}\big) = 18 - 3 = 15$.

The supremum (respectively, the infimum) of a set is also called its **least upper bound** (respectively, **greatest lower bound**).

Definition 1.5.22 (Maximum and minimum)**.** *Let A be a subset of $\mathbb{R}$ with at least one element. A number $\mathfrak{m}$ in $\mathbb{R}$ is said to be the **maximum** (resp. **minimum**) of A if*

i) $\mathfrak{m} = \sup A$ (*resp.* $\mathfrak{m} = \inf A$)*; and*

ii) $\mathfrak{m} \in A$.

We shall denote the maximum (resp. minimum) of A by $\max A$ (resp. $\min A$).

The maximum (resp. the minimum) of a subset of $\mathbb{R}$ is nothing but its greatest (resp. smallest) element, if such an element exists.

Example 1.5.23. It is clear that if S is a finite subset of $\mathbb{R}$ then $\inf S = \min S =$ smallest element of S and $\sup S = \max S =$ greatest element of S.

Example 1.5.24. Let $S = \{x \in \mathbb{R} : a < x < b\}$. Then $\sup S = b$ (resp. $\inf S = a$) but $b \notin S$ (resp. $a \notin S$), so S has no maximum (resp. minimum).

Example 1.5.25. Let $S = \{1/n : n \in \mathbb{N}\}$. Then $\sup S = 1$ (resp. $\inf S = 0$). Since $1 \in S$ (resp. $0 \notin S$) we have that $\max S = 1$ (resp. S has no minimum).

Example 1.5.26. Let $S = \{(-1)^n n/(2n+1) : n \in \mathbb{N}\}$. Then $\sup S = 1/2$ (resp. $\inf S = -1/2$) but $1/2 \notin S$ (resp. $-1/2 \notin S$), so S has no maximum (resp. minimum).

The fact that $\mathbb{N}$ has no supremum implies that for every real number r there is a natural number n such that $n > r$. A useful (an interesting[2]) consequence of this is the following fact about the distribution of the rational and irrational numbers in the real line.

Proposition 1.5.27. *Between any two distinct real numbers a and b there is always*

[2] Interesting, because, as you'll learn later on, there are many more irrational numbers than rationals.

i) *a rational number.*

ii) *an irrational number.*

Proof. i) Choose n in $\mathbb{N}$ big enough so that $n(b-a) > 1$. Then $nb-na > 1$ and $nb > na+1$. Hence there must be at least one integer m such that $na < m < nb$. Dividing by n, one obtains that $a < m/n < b$.

ii) By part (i), there is a rational number m/n such that $a/\sqrt{2} < m/n < b/\sqrt{2}$, and so, $a < \sqrt{2}\frac{m}{n} < b$. To finish, notice that $(m/n)\sqrt{2}$ must be irrational, for if $(m/n)\sqrt{2} = p/q$ for some integers p and q then $\sqrt{2} = \frac{pn}{qm}$, which is impossible. $\square$

Example 1.5.28. Let $S := \{q \in \mathbb{Q} : q^2 \leq 2\}$. Then $\sup S = \sqrt{2}$. Indeed,

(i) $q^2 \leq 2 \iff \sqrt{q^2} \leq \sqrt{2} \iff |q| \leq \sqrt{2} \iff -\sqrt{2} \leq q \leq \sqrt{2}$, so $\sqrt{2}$ is an upper bound.

(ii) Let $\varepsilon > 0$ arbitrary and let $c := \max\{-\sqrt{2}, \sqrt{2} - \varepsilon\}$. By Proposition 1.5.27, there exists $q \in \mathbb{Q}$ such that $c \leq q \leq \sqrt{2}$, so $q \in S$ and $\sqrt{2} - \varepsilon < q$.

Note that the last example shows that $\sqrt{2}$ is indeed a real number!

1.5.1 Decimal representations

To each real number one can associate a representation of the form $a_0 . a_1 a_2 a_3 \ldots a_n \ldots$, where a_0 is an integer and each a_i is a digit between 0 and 9. Such a representation is called the **decimal representation** of a real number, and it is this representation that we use in calculations. It can be obtained by successive approximations from below and from above as follows: Let's say we want to find the decimal representation of the real number r. First, one finds an integer a_0 such that $r \in [a_0, a_0 + 1)$. Then one divides the interval $[a_0, a_0+1)$ into 10 equal intervals, $[a_0.0, a_0.1)$, $[a_0.1, a_0.2)$, ..., $[a_0.9, a_0+1)$, and chooses the (only) one that contains r. The lower limit of this interval is $a_0.a_1$. Then, starting with the interval $[a_0.a_1, a_0.a_1 + 0.1)$, one repeats the same process. Continuing in this way, one can find any digit of the decimal representation of r.

It is possible to recognize a rational number out of its decimal representation, for decimal representations corresponding to rational numbers (and only those) have the property of being periodic, i.e., from a certain digit onwards each single digit of the representation will appear infinitely often in a periodic way. For instance, 1318/185 = 7.1243243243 ...

One last thing that we should mention about decimal representations is that some numbers can have more than one. For instance, $1.000\ldots = 0.999\ldots$.

We can also show that any decimal number is a real number. Let x be a decimal number $x = a_0.a_1a_2a_3\ldots a_n\ldots$, we can define

$$x_n = a_0.a_1a_2a_3\ldots a_n \in \mathbb{Q}, \quad \textit{and} \quad S(x) = \{x_n \ : \ n \in \mathbb{N}\}$$

It is clear that S is a set of non-empty real numbers, bounded about by $a_0 + 1$. The supremum of $S(x)$ is x, as $x - x_n < 10^{-n}$, so as long as $n > \log_{10}(1/\varepsilon)$ we have that $x - \varepsilon < x_n$.

1.6 The complex numbers ($\mathbb{C}$)

Why more numbers?

One could say that each extension, seen in previous sections, of a particular set of numbers was essentially motivated by the impossibility of solving some kind of equation within that set. For instance, the integers allowed us to solve equations like $x + 3 = 2$ (which couldn't be solved within the naturals); the rationals allowed us to solve equations like $2x = 3$ (which couldn't be solved within the integers), and the reals allowed us to solve equations like $x^2 = 2$ (which couldn't be solved within the rationals). It turns out that not every equation of the form

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0, \tag{1.3}$$

where $a_0, a_1, \ldots, a_n$ are all real numbers, can be solved within the reals[3]. For instance, an equation as simple as $x^2 + 1 = 0$ has no real solutions, because there is no real number whose square is -1. Problems of this kind led mathematicians, in the 16th century, to the invention of the complex numbers – a larger set, containing the reals and the number $\sqrt{-1}$. Later on, in the 18th century, it was shown that this new set of numbers was the ultimate one, in the sense that every equation like (1.3) with $a_0, a_1, \ldots, a_n$ complex has a complex solution. In this section, we have collected some basic but important facts about the complex numbers. Later on, you will learn more about them.

First of all, a **complex number** is a number of the form $a + ib$, where a and b are real numbers and i is a mathematical symbol called the imaginary unit. The real number a is called the **real part** of the complex number and denoted $\mathrm{Re}(a + ib)$; and the real number b is called the **imaginary part** and denoted $\mathrm{Im}(a + ib)$. Thus, every real number is a complex number, for if a is real, then it can be written as $a = a + i0$. We shall denote the collection of all complex numbers by $\mathbb{C}$.

Complex numbers are **added and multiplied** according to the following rules:

$$(a + ib) + (c + id) = (a + c) + i(b + d),$$

and

$$(a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc).$$

One can check that, with addition and multiplication defined in this way, the complex numbers obey the same basic laws of arithmetic as reals. Moreover, notice that $i^2 = -1$ (or equivalently, $i = \sqrt{-1}$), so the equation $x^2 + 1 = 0$ has at least one complex solution, namely i.

Example 1.6.1. More generally, any equation of the form $ax^2 + bx + c = 0$ has solutions:

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \qquad \text{and} \qquad x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$ [4]

So, for instance, the solutions of the equation $x^2 + 2x + 5 = 0$ are $x_1 = \dfrac{-2 + \sqrt{4 - 4 \cdot 5}}{2} = -1 + \sqrt{1 - 5} = -1 + 2i$ and $x_2 = \dfrac{-2 - \sqrt{4 - 4 \cdot 5}}{2} = -1 - \sqrt{1 - 5} = -1 - 2i$.

[3] An equation of this form is called a polynomial equation and the expression figuring on its left hand side is called, as you surely know, a polynomial (whence the name).

[4] Of course, they will coincide if $b^2 - 4ac = 0$.

Two complex numbers are **equal** if they have the same real and imaginary parts. Every complex number $a + ib \neq 0$ has an **inverse**, namely, the number

$$\frac{a}{a^2+b^2} - i\frac{b}{a^2+b^2}$$

(here, like with the rationals, we say that a complex number β is the inverse of a complex number α if $\alpha \cdot \beta = 1$).

Given a complex number $\alpha = a + ib$, its **conjugate**, denoted $\overline{\alpha}$, is defined to be the number $a - ib$. Here are some useful properties of the conjugate.

Proposition 1.6.2. *For every $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{C}$, one has that $\overline{\alpha_1 \cdot \alpha_2 \cdot \ldots \cdot \alpha_n} = \overline{\alpha}_1 \cdot \overline{\alpha}_2 \cdot \ldots \cdot \overline{\alpha}_n$ and $\overline{\alpha_1 + \alpha_2 + \ldots + \alpha_n} = \overline{\alpha}_1 + \overline{\alpha}_2 + \ldots + \overline{\alpha}_n$. Moreover, $\overline{\overline{\alpha}} = \alpha$ for every $\alpha \in \mathbb{C}$.*

The **absolute value** or **modulus** of a complex number $\alpha = a + ib$ is defined as

$$|\alpha| = \sqrt{a^2+b^2}.$$

The conjugate and modulus of a complex number α are related by the identity

$$|\alpha|^2 = \alpha\overline{\alpha}.$$

Complex numbers can be naturally identified with the points of the **cartesian plane**[5]. Precisely, one identifies the complex number $a + ib$ with the point (a, b). Accordingly, the modulus of $a + ib$ is nothing but the distance between the point (a, b) and the origin and the conjugate of $a + ib$ corresponds to the point $(a, -b)$ symmetric to (a, b) with respect to the x-axis. Furthermore, if a and b are the coordinates of a point in the cartesian plane then

$$a = r\cos\theta \qquad \text{and} \qquad b = r\sin\theta,$$

where $r = \sqrt{a^2+b^2} = |a + ib|$ and θ is the angle between the positive part of the x-axis and the segment joining the origin with the point (a, b), so $a + ib$ can be written as

$$re^{i\theta} = r(\cos\theta + i\sin\theta).$$

This last expression is called the **polar form** of the complex number $a+ib$. The angle θ is called 'the' **argument** of the complex number $a + ib$ and denoted $\arg(a + ib)$. Clearly, θ is defined only to within a multiple of 2π. However, there is only one value of θ satisfying

$$-\pi < \theta \leq \pi.$$

We shall call this value of θ the **principal value of the argument** of the complex number $a + ib$ and denote it by $\mathrm{Arg}(a + ib)$. We sometimes write (r, θ) for a complex number in polar form, that is,

$$(r, \theta) = r(\cos\theta + i\sin\theta).$$

[5]Recall, the cartesian plane is nothing but a pair of oriented perpendicular straight lines (called the x and y axes) along which distances to their point of intersection (called the origin) are measured using the same length unit. The position of a point in it is then specified by its coordinates, i.e., the distances between the origin and the perpendicular projections of the point on the x and y axes (taken in this order).

Like it happens with the modulus and the conjugate, also multiplication of complex numbers has a very simple geometrical meaning, which gets revealed immediately if one writes down the formula for multiplication in polar form. Precisely, we have that

$$\begin{aligned}
&r_1(\cos\theta_1 + i\sin\theta_1)\cdot r_2(\cos\theta_2 + i\sin\theta_2)\\
&\qquad = r_1r_2\big(\cos\theta_1\cos\theta_2 + i\cos\theta_1\sin\theta_2 + i\sin\theta_1\cos\theta_2 + i^2\sin\theta_1\sin\theta_2\big)\\
&\qquad = r_1r_2\big[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\cos\theta_1\sin\theta_2 + \sin\theta_1\cos\theta_2)\big]\\
&\qquad = r_1r_2\big(\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\big).
\end{aligned}$$

So multiplication by a complex number of modulus r and argument θ amounts, in geometric terms, to a dilation by r combined with a rotation by the angle θ about the origin.

In fact the formula

$$(r,\theta)\cdot(s,\phi) = (rs, \theta+\phi)$$

is easy to remember and writing it out gives you the double angle formula for both sine and cosine. This is how I remember the double angle formula. To justify $e^{i\theta} = \cos(\theta) + i\sin(\theta)$, we can argue as follows

$$\begin{array}{ll}
e^x = 1 + x + x^2/2 + x^3/3! + x^4/4! + \ldots & e^{ix} = 1 + ix - x^2/2 - ix^3/3! + x^4/4! + \ldots\\
\sin(x) = x - x^3/3! + x^5/5! - \ldots & i\sin(x) = ix - ix^3/3! + ix^5/5! - \ldots\\
\cos(x) = 1 - x^2/2 + x^4/4! - \ldots &
\end{array}$$

Another pattern is the following way of remembering the standard trig values.

	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
sin	$\sqrt{0}/2$	$\sqrt{1}/2$	$\sqrt{2}/2 = 1/\sqrt{2}$	$\sqrt{3}/2$	$\sqrt{4}/2 = 1$
cos	$\sqrt{4}/2 = 1$	$\sqrt{3}/2$	$\sqrt{2}/2 = 1/\sqrt{2}$	$\sqrt{1}/2$	$\sqrt{0}/2$

A simple but useful consequence of the polar form of the multiplication formula is the following result known as **de Moivre's Formula**. It states that for any real number θ and every natural number n one has that

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta.$$

The formula can be easily proved by induction on n. Indeed, it obviously holds true for $n = 1$, and if it holds true for some $n \geq 1$ then

$$\begin{aligned}
(\cos\theta + i\sin\theta)^{n+1} &= (\cos\theta + i\sin\theta)^n(\cos\theta + i\sin\theta)\\
&= (\cos n\theta + i\sin n\theta)(\cos\theta + i\sin\theta)\\
&= \cos{(n+1)\theta} + i\sin{(n+1)\theta},
\end{aligned}$$

where the last equality follows from the polar form of the formula for the product of two complex numbers. This can also be seen as $(e^{i\theta})^n = e^{in\theta}$.

De Moivre's Formula makes it easy to derive expressions for $\cos nx$ and $\sin nx$ in terms of $\cos x$ and $\sin x$. However, in doing this one also needs the binomial theorem, which shall be discussed in the next chapter. We thus postpone until then the derivation of such expressions.

1.6.1 Something for extra reading

Through de Moivre's Formula it is also possible to find explicit formulas for the complex solutions of equations of the form $z^n - \alpha = 0$, where $\alpha \in \mathbb{C}$. Indeed, letting $z = r(\cos\theta + i\sin\theta)$ and writing α as $|\alpha|\big(\cos(\mathrm{Arg}(\alpha)) + i\sin(\mathrm{Arg}(\alpha))\big)$, we obtain that

$$r^n(\cos n\theta + i\sin n\theta) = |\alpha|\big(\cos(\mathrm{Arg}(\alpha)) + i\sin(\mathrm{Arg}(\alpha))\big),$$

which is possible if and only if $r^n = |\alpha|$ and $n\theta = \mathrm{Arg}(\alpha) + 2k\pi$ $(k \in \mathbb{Z})$. From this, we conclude that $r = \sqrt[n]{|\alpha|}$ and $\theta = (\mathrm{Arg}(\alpha) + 2k\pi)/n$. Thus, each number of the form

$$\sqrt[n]{|\alpha|}\left(\cos\left(\frac{\mathrm{Arg}(\alpha) + 2k\pi}{n}\right) + i\sin\left(\frac{\mathrm{Arg}(\alpha) + 2k\pi}{n}\right)\right) \qquad (k \in \mathbb{Z}), \tag{1.4}$$

is a solution of the equation $z^n - \alpha = 0$. At first glance, it may seem from (1.4) as if the equation had infinitely many solutions. However, if one represents these solutions in the complex plane, starting with the one corresponding to $k = 0$, it becomes apparent that, for each value of k between 0 and $n-1$ one obtains a different complex number, but as k becomes either negative or greater than $n-1$ any 'new' solution given by (1.4) will coincide with one of the n solutions already listed. So the list of all distinct (complex) solutions of the equation $z^n - \alpha = 0$ is:

$$\sqrt[n]{|\alpha|}\left(\cos\left(\frac{\mathrm{Arg}(\alpha) + 2k\pi}{n}\right) + i\sin\left(\frac{\mathrm{Arg}(\alpha) + 2k\pi}{n}\right)\right) \qquad (0 \le k < n).$$

When $\alpha = 1$ one has that $|\alpha| = 1$ and $\mathrm{Arg}(\alpha) = 0$, so the (complex) solutions of the equation $z^n - 1 = 0$ take the form

$$\cos\left(\frac{2k\pi}{n}\right) + i\sin\left(\frac{2k\pi}{n}\right) \qquad (0 \le k < n). \tag{1.5}$$

These numbers are known (for the obvious reasons) as the **n-th roots of unity**.

Example 1.6.3. Find all solutions of the equation $z^3 - 1 = 0$ in $\mathbb{C}$.

According to (1.5), the solutions of $z^3 - 1 = 0$ are:

$$\begin{aligned}
\cos\left(\frac{0\pi}{3}\right) + i\sin\left(\frac{0\pi}{3}\right) &= 1 + i\,0 = 1,\\
\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right) &= -\frac{1}{2} + i\frac{\sqrt{3}}{2},\\
\cos\left(\frac{4\pi}{3}\right) + i\sin\left(\frac{4\pi}{3}\right) &= -\frac{1}{2} - i\frac{\sqrt{3}}{2}.
\end{aligned}$$

Chapter 2

Sets and functions

In this chapter, we return to the language of sets. We introduce important ways of combining sets (intersection, union, product) and give the abstract definition of the already familiar concept of a function.

2.1 The algebra of sets

In this section, we introduce several important operations between sets and look at the rules governing them. These operations could be seen as set-theoretic analogues of the arithmetic operations and order relations between numbers studied in the previous chapter. Our starting point is to look at what it means for sets to be equal.

We recall that a set is a collection of mathematical objects, where repeats do not matter and there is no ordering. We are happy that a set can be written in different ways. For example the set of odd numbers can be written in the following four ways:

$$\begin{array}{ll} \{1,3,5,7,\dots\} & \{2a-1 \ : \ a \in \mathbb{N}\} \\ \{k \in \mathbb{N} \ : \ k \text{ is odd}\} & \{z \in \mathbb{N} \ : \ z = 2f+1 \text{ for some } f \in \mathbb{Z}\}. \end{array}$$

We know informally that these four sets agree, but how can we write this in a formal mathematical language? There are many sets that we can describe, but not be able to write down any elements

$$\{a \in \mathbb{N} \ : \ a \text{ is even, bigger than 2 and is not the sum of two primes.}\}$$

Definition 2.1.1 (Relations between sets)**.** *Given sets A and B, we shall say that A is a* ***subset*** *of B, denoted $A \subseteq B$ (or $B \supseteq A$), if every element of A is also an element of B. We shall say that A and B are* ***equal****, written $A = B$, if they contain the same elements. If $A \subseteq B$ and $A \neq B$ then we shall say that A is a* ***proper subset*** *of B. We shall often write this last as $A \subset B$ (or $B \supset A$).*

Remark 2.1.2. It follows readily from this last definition that two sets A and B are equal if and only if $A \subseteq B$ and $B \subseteq A$.

This is how we decide two sets are the same, we check that each is contained in the other.

Example 2.1.3. Let $A := \{n \in \mathbb{Z} : (n{+}5)/2 \in \mathbb{N}\}$ and $B := \{n \in \mathbb{Z} : -n \leq 3 \text{ and } n \text{ is odd}\}$. Show that $A = B$.

$A \subseteq B$: Let $n \in A$ arbitrary. $(n+5)/2 \in \mathbb{N} \Rightarrow (n+5)/2 \geq 1 \Rightarrow n+5 \geq 2 \Rightarrow n \geq -3$, so $-n \leq 3$. Next, set $(n+5)/2 = k$. Then $n = 2k-5 = 2(k-3)+1$, which is clearly odd. Thus $n \in B$, and since this holds for any $n \in A$, $A \subseteq B$.

$A \supseteq B$: Let $n \in B$ arbitrary. First, $-n \leq 3 \Rightarrow n \geq -3 \Rightarrow n+5 \geq 2 \Rightarrow (n+5)/2 \geq 1$. Also, n odd $\Rightarrow n = 2k-1$ for some $k \in \mathbb{Z} \Rightarrow (n+5)/2 = [(2k-1)+5]/2 = k+2 \in \mathbb{Z}$. Thus, $(n+5)/2 \in \mathbb{N}$. This shows $n \in A$, and as this holds for any $n \in B$, $B \subseteq A$.

Lastly, since $A \subseteq B$ and $B \subseteq A$, $A = B$.

It may happen that, while solving a mathematical problem, all the objects under consideration belong to a certain set. This set is usually referred to as a **universal set** for the problem.

Definition 2.1.4. *Let A and B be arbitrary sets.*

- *The **union** of A and B, denoted $A \cup B$, is the set of all those elements that belong to either A or B, i.e., $A \cup B = \big\{x : x \in A \text{ or } x \in B\big\}$.*

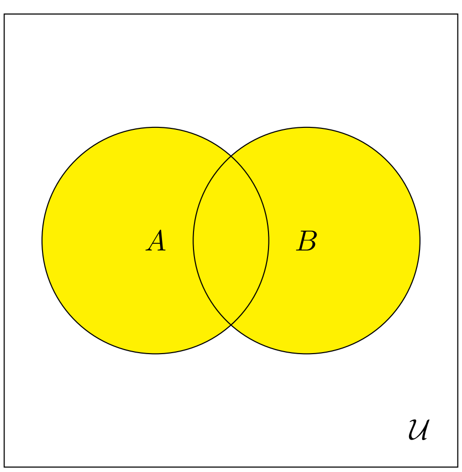
- *The **intersection** of A and B, denoted $A \cap B$, is the set of all those elements that belong to both A and B, i.e., $A \cap B = \big\{x : x \in A \text{ and } x \in B\big\}$.*

- *The **difference** of A and B, denoted $A \setminus B$, is the set of all those elements that belong to A but do not belong to B, i.e., $A \setminus B = \big\{x : x \in A \text{ and } x \notin B\big\}$.*

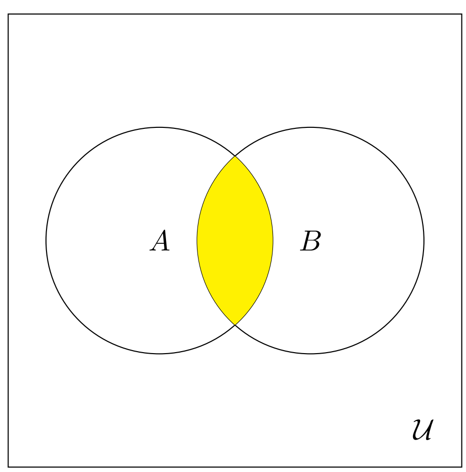
- *If A is a subset of a universal set $\mathcal{U}$ then the **complement** of A (in $\mathcal{U}$), denoted A^c, is the set $\mathcal{U} \setminus A$.*

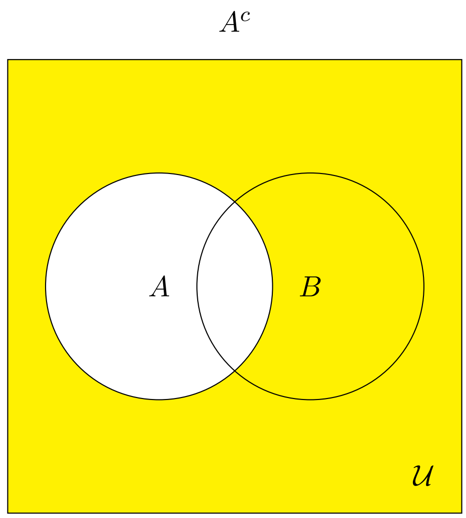
We can draw these various sets in terms of Venn diagrams. In each case we imagine there is some sensible universal set $\mathcal{U}$ that contains both A and B, this is drawn as the outside rectangle.

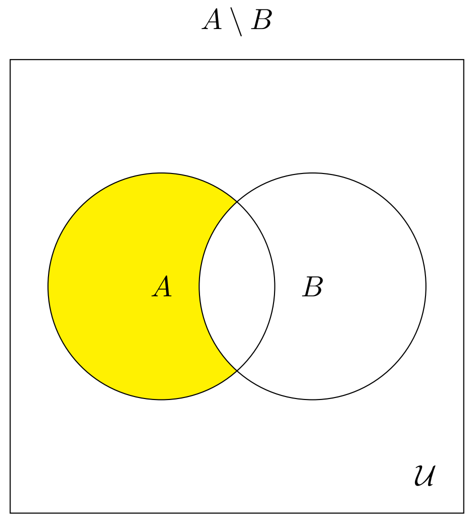
$A \cup B$



$A \cap B$







Example 2.1.5. Let $A := \{n \in \mathbb{N} : n < 11\}$, $B := \{z \in \mathbb{Z} : -10 < z < 5\}$ and $C := \{q \in \mathbb{Q} : q < 0 \text{ or } q \text{ is an even natural number}\}$. Then

- $A \cup B = \{z \in \mathbb{Z} : -10 < z < 11\}$.
- $A \cap B = \{n \in \mathbb{N} : 1 \leq n < 5\}$ and $C \cap B = \{z \in \mathbb{Z} : -10 < z < 0 \text{ or } z = 2 \text{ or } z = 4\}$.
- $A \setminus B = \{z \in \mathbb{Z} : 5 \leq z < 11\}$ and $B \setminus C = \{0, 1, 3\}$.

Venn diagrams are not a way to prove something! They are just a helpful visualisation. The amount you'd have to write to explain the diagrams is much more than in a logical argument. Instead we argue using the language of sets.

Example 2.1.6. Show that for any three sets A, B and C one has that $(A \subseteq C \text{ and } B \subseteq C) \Rightarrow A \cup B \subseteq C$.
Solution: $x \in A \cup B \iff (x \in A \text{ or } x \in B) \Rightarrow x \in C$.

There is a set that plays with respect to the above operations a similar role to that played by the number 0 within the number systems. It is formally defined to be the set with no elements. It is called the **empty set** and denoted by $\emptyset$.

Our next proposition contains some obvious but very useful relations involving the above operations. We omit its proof.

Proposition 2.1.7. *Let A be any set. Then*

i) $A \cup \emptyset = A$.

ii) $A \cap \emptyset = \emptyset$.

iii) $A \cup A = A$.

iv) $A \cap A = A$.

Moreover, if A is a subset of a universal set $\mathcal{U}$ then

v) $A \cap A^c = \emptyset$.

vi) $A \cup A^c = \mathcal{U}$.

vii) $A \cup \mathcal{U} = \mathcal{U}$.

viii) $A \cap \mathcal{U} = A$.

ix) $(A^c)^c = A$.

Less obvious but far more important and useful are the relations contained in the next two theorems.

Theorem 2.1.8 (Distributive Laws). *Let A, B and C be arbitrary sets. Then*

i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Proof. i) $x \in A \cap (B \cup C) \iff x \in A$ and $x \in B \cup C \iff x \in A$ and $\big(x \in B$ or $x \in C\big)$ $\iff \big(x \in A$ and $x \in B\big)$ or $\big(x \in A$ and $x \in C\big) \iff x \in A \cap B$ or $x \in A \cap C \iff$ $x \in (A \cap B) \cup (A \cap C)$.

ii) See Example 2.1.10 below. □

Theorem 2.1.9 (De Morgan's Laws). *Let A and B be subsets of a universal set $\mathcal{U}$. Then*

i) $(A \cup B)^c = A^c \cap B^c$.

ii) $(A \cap B)^c = A^c \cup B^c$.

Proof. i) $x \in (A \cup B)^c \iff x \notin A \cup B \iff x \notin A$ and $x \notin B \iff x \in A^c$ and $x \in B^c$ $\iff x \in A^c \cap B^c$.

ii) Tutorial exercise. □

Example 2.1.10. Let A, B and C be arbitrary subsets of a universal set $\mathcal{U}$. Then, applying Theorem 2.1.9 and part (i) of Theorem 2.1.8, we find that

$$\begin{aligned}[A \cup (B \cap C)]^c &= A^c \cap (B \cap C)^c \;=\; A^c \cap (B^c \cup C^c) \;=\; (A^c \cap B^c) \cup (A^c \cap C^c) \\ &= (A \cup B)^c \cup (A \cup C)^c \;=\; [(A \cup B) \cap (A \cup C)]^c.\end{aligned}$$

It follows that

$$A \cup (B \cap C) = \big[[A \cup (B \cap C)]^c\big]^c = \big[[(A \cup B) \cap (A \cup C)]^c\big]^c = (A \cup B) \cap (A \cup C)$$

(see Proposition 2.1.7 (vi)). This proves part (ii) of Theorem 2.1.8[1].

[1] Note that the assumption that A, B and C are subsets of a universal set $\mathcal{U}$ (which is not amongst the hypotheses of Theorem 2.1.8 (ii)) is not a significant constraint since, given A, B and C, we can always take as universal set $A \cup B \cup C$.

Example 2.1.11. Assuming known that, for any pair of subsets E and F of some universal set $\mathcal{U}$, the identity $E \setminus F = F^c \cap E$ holds (this identity will be proved in a tutorial), show that for any three subsets A, B and C of $\mathcal{U}$,

$$(B \setminus A) \cup C = (B \cup C) \setminus (A \setminus C).$$

Combining the given identity with those from Proposition 2.1.7 (vi), Theorem 2.1.8 (ii) and Theorem 2.1.9 (ii), we find that

$$\begin{aligned} (B \setminus A) \cup C &= (A^c \cap B) \cup C \;=\; (A^c \cup C) \cap (B \cup C) \;=\; (A \cap C^c)^c \cap (B \cup C) \\ &= (B \cup C) \setminus (A \cap C^c) \;=\; (B \cup C) \setminus (A \setminus C). \end{aligned}$$

2.1.1 Aside on infinite sets

The above can seem quite simple with finite sets, but I want to point out how weird infinity can be. I want to give a mangled version of a 1924 lecture by David Hilbert.

Imagine a hotel with one room for each $n \in \mathbb{N}$ and each room is occupied.

What can we do when a new guest arrives? We put the guest into room 1, then move the person of room 1 into room 2 and so on. Each guest only moves once, so we won't annoy them all.

Now a coach arrives with one passenger for each $n \in \mathbb{N}$. Can we accommodate them? We put the guest of room 1 into room 2, the guest of room 2 into room 4, and so on. We then put the first passenger into room 1, the second into room 3, and so on. Again each guest only moves once.

Guest	Passenger
$1 \mapsto 2$	$1 \mapsto 1$
$2 \mapsto 4$	$2 \mapsto 3$
$3 \mapsto 6$	$3 \mapsto 5$
$\vdots$	$\vdots$
$k \mapsto 2k$	$a \mapsto 2a-1$

Now one coach arrives for each $n \in \mathbb{N}$. Can we accommodate them?

Guest	Coach 1	Coach 2	$\dots$
$1 \mapsto 2$	$1 \mapsto 3$	$1 \mapsto 5$	$\dots$
$2 \mapsto 4$	$2 \mapsto 3^2$	$2 \mapsto 5^2$	$\dots$
$3 \mapsto 6$	$3 \mapsto 3^3$	$3 \mapsto 5^3$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$k \mapsto 2k$	$a \mapsto 3^a$	$a \mapsto 5^a$	$\dots$

In general passenger k from coach m is sent to room p_m^k, where p_m is the mth odd prime number.

The point is that infinity is weird, so that "all rooms occupied" is not the same as "cannot accommodate any more guests".

But if a ferry arrives with one passenger for each $x \in \mathbb{R}$, we cannot accommodate them all! This comes from the fact that $\mathbb{N}$ is countable and $\mathbb{R}$ is not. This is all made clear in level three set theory.

2.1.2 The cartesian product

There is another 'operation' between sets, which is fairly common and useful, and that extends the idea of the cartesian plane that you probably learned at school.

Definition 2.1.12. *Given sets $S_1, S_2, \ldots, S_k$, we define their (cartesian)* ***product****, denoted $S_1 \times S_2 \times \cdots \times S_k$, to be the set of all ordered k-tuples $(s_1, s_2, \ldots, s_k)$ such that $s_i \in S_i$ for every $1 \leq i \leq k$.*

Example 2.1.13.

- Let $X = \{1, 2, 3\}$ and $Y = \{a, b\}$. Then $X \times Y = \{(1, a), (2, a), (3, a), (1, b), (2, b), (3, b)\}$.
- Let $X = Y = \mathbb{R}$. Then $X \times Y = \{(x, y) : x, y \in \mathbb{R}\}$. This set can be identified with the cartesian plane. Precisely, one can associate to each point in the cartesian plane the ordered pair consisting of its coordinates. Likewise, if one has a third set, Z say, also equal to $\mathbb{R}$, then $X \times Y \times Z$ can be (in the same way) identified with the points of three-dimensional space.
- Let $X = Y = \mathbb{Z}$. Then, in view of the previous example, one can think of $X \times Y = \{(x, y) : x, y \in \mathbb{Z}\}$ as the lattice formed by all points in the cartesian plane with integer coordinates.

2.1.3 Unions and intersections of any number of sets

Intersections and unions can be extended to any number (possibly infinite) of sets. For example, let us give the definitions in the case of a finite family of sets.

Given a family of sets $A_1, A_2, \ldots, A_n$,

- Their union, denoted $A_1 \cup A_2 \cup \cdots \cup A_n$ or $\bigcup_{i=1}^n A_i$, is the set of all those elements that belong to at least one of the sets in the family.
- Their intersection, denoted $A_1 \cap A_2 \cap \cdots \cap A_n$ or $\bigcap_{i=1}^n A_i$, is the set of all those elements that belong to each one of the sets in the family.

We won't make much use of these extended notions in this course but they will become very important and common in the future.

2.2 Some problems of enumerative combinatorics (a small digression)

Fairly often in mathematics, we come across problems in which one needs to determine the number of ways in which certain patterns can be formed out of a given collection of objects. Problems of this kind are known as combinatorial problems. This section is devoted to them.

The three problems that follow are amongst the most common combinatorial problems that you will encounter in your studies.

Problem 1: *Given sets $S_1, S_2, \ldots, S_k$ with $n_1, n_2, \ldots, n_k$ elements, respectively, how many k-tuples are there in $S_1 \times \ldots \times S_k$?*

Answer: $n_1 \cdot n_2 \cdot \ldots \cdot n_k$.

Sketch of the proof: There are n_1 possible ways of choosing the first entry of a k-tuple in $S_1 \times \ldots \times S_k$. Then, for each choice of the first entry there are n_2 possible ways of choosing the second entry. Then, for each choice of the first and second entries, there are n_3 possible ways of choosing the third entry. Continuing in this way, one arrives at the answer.

Problem 2: *Given a set S with n elements, how many k-tuples without repeated entries are there in $\underbrace{S \times S \times \cdots \times S}_{k \text{ times}}$?*

Answer: $n(n-1)\ldots(n-k+1)$, or equivalently, $\dfrac{n!}{(n-k)!}$ (recall that $0! = 1! = 1$ and, for every $n > 1$, $n! = (n-1)!\,n$).

Sketch of the proof: There are n possible ways of choosing the first entry of a k-tuple in the product of k copies of S. But, for each choice of the first entry, there are only $n-1$ possible ways of choosing the second entry (recall this time, entries cannot be repeated). Then, for each choice of the first and second entries, there are $n-2$ possible ways of choosing the third entry. Continuing in this way, one arrives at the answer.

A special case of Problem 2 arises when $k = n$. In this case, the question is equivalent to the following: *In how many different ways can we arrange n different objects in a row?* Such arrangements are often referred to as **permutations**. Of course, the answer is $n!$.

Problem 3: *Given a set S with n elements, how many different subsets of k elements does S have? (Of course, $k \leq n$.)*

Answer: $\dfrac{n!}{k!(n-k)!} =: \binom{n}{k}$.

Proof: There are $n!/(n-k)!$ possible ways of choosing an ordered k-tuple without repeated entries from a set with n elements. On the other hand, given a set of k elements, there are $k!$ possible ways of arranging them to form an ordered k-tuple. Thus, the total number of subsets of S with k elements is $\dfrac{n!}{k!(n-k)!}$.

The numbers $\binom{n}{k}$ are called the **binomial coefficients** (below, we will see why) and, as you probably know, they are the numbers that appear in the rows of Pascal's triangle. They satisfy the identity $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$ for every $1 \leq k \leq n$[1], which is something you know – recall how the rows of Pascal's triangle are constructed!

In practice, one may need to combine the results above in order to solve a given

[1]Indeed,

$$\begin{aligned}
\binom{n}{k-1} + \binom{n}{k} &= \frac{n!}{(k-1)!(n-k+1)!} + \frac{n!}{k!(n-k)!} = \frac{n!}{(k-1)!(n-k)!}\left(\frac{1}{n-k+1} + \frac{1}{k}\right) \\
&= \frac{n!}{(k-1)!(n-k)!} \cdot \frac{n+1}{(n-k+1)k} = \frac{(n+1)!}{k!(n+1-k)!} = \binom{n+1}{k}.
\end{aligned}$$

problem. Let's see this through concrete examples.

Example 2.2.1. Five friends eat lunch on a long bench every afternoon. How many days in a row can they eat lunch without sitting in the same order?

There are 5! possible arrangements of the five friends in the bench. Hence, 5! days is the answer.

Example 2.2.2. How many different triangles can be formed using the vertices of a regular n-sided polygon?

Since three distinct points determine a triangle, the question essentially asks for the number of different ways in which one can choose three points from a set of n points. Of course, the answer is $\binom{n}{3}$.

Remark 2.2.3. I want to make clear that simply giving the correct number isn't worth much and is worth nothing if you get it wrong! Instead, you should give a (brief) argument as to why how you have reasoned to your answer. This means that you can get partial marks if you make a small mistake.

Example 2.2.4. a) How many teams of 5 players can be chosen from a group of 10 players?

b) How many teams will include the best player and exclude the worst player?

a) Clearly, $\binom{10}{5}$.

b) If the best player must be in the team then we only need to choose 4 players. In addition, as the worst player cannot be in the team, we won't be choosing from 9 players but only from 8. So, this time, the answer is $\binom{8}{4}$.

Example 2.2.5. Four different dice are rolled.

a) In how many outcomes will at least one five appear?

b) In how many outcomes will the highest die be a five?

a) The number of possible outcomes is clearly 6^4. Of them, only 5^4 do not contain 5. Thus, the total number of outcomes in which at least one 5 appears is $6^4 - 5^4$.

b) This time 6 cannot appear in any of the outcomes. Clearly, there are 5^4 possible outcomes with this property, of which 4^4 do not contain 5. Thus, the number of possibilities is $5^4 - 4^4$.

Example 2.2.6. Three different math books and two different physics books are arranged on a bookshelf.

a) In how many different orders can they be arranged if all the math books are together and all the physics books are together?

b) In how many ways can they be arranged if no two math books are together?

a) We can either put the physics books first and the math books second or the other way around. At the same time there are 3! possible ways to arrange the 3 math books and 2! ways to arrange the 2 physics books. Thus, altogether, there are $2 \times (3! \times 2!)$ arrangements.

b) Since there are 3 math books and only 2 physics books there is only one way not to have 2 math books together, and it is by placing the physics books in an alternating fashion between the math books. Now, there are 2! ways of ordering the physics books amongst the maths books and 3! ways of ordering the math books, so the answer to the question is $2! \times 3!$.

Example 2.2.7. On the menu of a Chinese restaurant there are 7 chicken dishes, 6 beef dishes, 6 pork dishes, 8 seafood dishes, and 9 vegetable dishes.

a) In how many ways can a family order if they choose exactly one dish of each kind?

b) In how many ways can they order if at most one dish of each kind is chosen?

a) Clearly, the answer is $7 \times 6 \times 6 \times 8 \times 9$.

b) There are 7 possible ways of choosing a chicken dish. But we also need to take into account the possibility that no chicken dish is chosen. So there are really 8 possible choices when it comes to choose a chicken dish. Likewise, there are 7 possible choices when it comes to choose a beef dish; 7 possible choices when it comes to choose a pork dish; 9 possible choices when it comes to choose a seafood dish; and 10 possible choices when it comes to choose a vegetable dish. Thus, there are in total $8 \times 7 \times 7 \times 9 \times 10 - 1$ ways to order a meal, where the -1 comes from the fact that the 'empty order' is not an order.

Example 2.2.8. Six high school bands are lined up for a parade.

a) In how many orders can the six bands be lined up if two of the bands have red uniforms and must be separated by at least one band?

b) In how many ways can they be lined up if the two bands with red uniforms are put at opposite ends of the parade?

a) Place first the 4 bands with uniforms other than red. There are $\binom{5}{2}$ ways of choosing two places for the red uniform bands amongst the others. In addition, there are 2! ways of arranging the red uniform bands and 4! ways of arranging the other four. Thus, there $4! \times 2! \times \binom{5}{2}$ possible line-ups for the bands.

b) In this case, the only freedom we have is to arrange the red uniform bands and the group of the other four. The answer is thus $2! \times 4!$.

Example 2.2.9. Eight different pairs of shoes are stored in a bag. If two right shoes and two left shoes are chosen at random, what is the probability that at least one matching pair of shoes is chosen?

There are in total $\binom{8}{2} \times \binom{8}{2}$ possible ways of choosing 2 right shoes and 2 left shoes from the bag, of which, $\binom{8}{2} \times \binom{6}{2}$ do not yield a matching pair. Thus, the number of ways of choosing 2 right shoes and 2 left shoes from the bag so that at least one matching pair is chosen is $\binom{8}{2} \times \binom{8}{2} - \binom{8}{2} \times \binom{6}{2}$. The required probability is then $\left(\binom{8}{2} \times \binom{8}{2} - \binom{8}{2} \times \binom{6}{2}\right)/\left(\binom{8}{2} \times \binom{8}{2}\right) = 1 - \binom{6}{2}/\binom{8}{2}$.

The last result of this section, although algebraic in nature, it is closely related to Problem 3 above. Before stating it, let us introduce a useful notation. Given a sum $x_k + x_{k+1} + x_{k+2} + \cdots + x_n$ we will write it down as

$$x_k + x_{k+1} + x_{k+2} + \cdots + x_n = \sum_{i=k}^{n} x_i,$$

to be read *"summation from i equal k to n of x_i"*. Similarly we can write

$$x_k x_{k+1} x_{k+2} \cdots x_n = \Pi_{i=k}^{n} x_i,$$

to be read *"the product of x_i from i equals k to n"*.

Theorem 2.2.10 (Binomial Theorem)**.** *For any pair of numbers x and y and any natural number n one has that*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Proof. We have $(x+y)(x+y)\ldots(x+y)$, n copies. When expanding this term at each bracket we pick either x or y. If we choose k copies of x we get $x^k y^{n-k}$. Since the ordering doesn't matter, there are $\binom{n}{k}$ ways of getting this term. The result follows. □

Proof. The proof is by induction on n.

For $n = 1$ the result is clearly true.

Suppose the formula has been established for some natural number $n \geq 1$. Then we have that

$$\begin{aligned}
(x+y)^{n+1} &= (x+y)(x+y)^n = (x+y)\sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k} \\
&= \sum_{k=0}^{n}\binom{n}{k}x^{k+1}y^{n-k} + \sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k+1} \\
&= \sum_{k=0}^{n}\binom{n}{k}x^{k+1}y^{(n+1)-(k+1)} + \sum_{k=0}^{n}\binom{n}{k}x^k y^{(n+1)-k} \\
&= \sum_{k=1}^{n+1}\binom{n}{k-1}x^k y^{(n+1)-k} + \sum_{k=0}^{n}\binom{n}{k}x^k y^{(n+1)-k} \\
&= \binom{n}{0}y^{n+1} + \sum_{k=1}^{n}\left[\binom{n}{k-1} + \binom{n}{k}\right]x^k y^{(n+1)-k} + \binom{n}{n}x^{n+1} \\
&= \binom{n+1}{0}x^0 y^{n+1} + \sum_{k=1}^{n}\binom{n+1}{k}x^k y^{(n+1)-k} + \binom{n+1}{n+1}x^{n+1}y^0 \\
&= \sum_{k=0}^{n+1}\binom{n+1}{k}x^k y^{(n+1)-k}.
\end{aligned}$$

This shows the formula holds for $n+1$.

By mathematical induction, one concludes that the formula holds true for every n. □

Example 2.2.11. How many subsets (of any size) does a set S of n elements have?

Since the number of subsets of S with k elements is $\binom{n}{k}$, the total number of subsets of S is

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n}.$$

Letting $x = y = 1$ in the binomial formula, we obtain that the value of this last sum is 2^n.

Example 2.2.12. Combining de Moivre's and the binomial formulas, one can easily find expressions for $\cos n\theta$ and $\sin n\theta$ in terms of $\cos\theta$ and $\sin\theta$. For instance, let $n = 3$ and let $\theta \in \mathbb{R}$ arbitrary. Then

$$\begin{aligned}
\cos 3\theta + i\sin 3\theta &= (\cos\theta + i\sin\theta)^3 \\
&= \binom{3}{0}\cos^3\theta + \binom{3}{1} i\sin\theta\cos^2\theta + \binom{3}{2} i^2\sin^2\theta\cos\theta + \binom{3}{3} i^3\sin^3\theta \\
&= \cos^3\theta + 3i\sin\theta\cos^2\theta - 3\sin^2\theta\cos\theta - i\sin^3\theta \\
&= \cos^3\theta - 3\sin^2\theta\cos\theta + i(3\sin\theta\cos^2\theta - \sin^3\theta),
\end{aligned}$$

and equating real parts, one obtains the formula for the cosine of the triple angle:

$$\cos 3\theta = \cos^3\theta - 3\sin^2\theta\cos\theta.$$

2.3 Functions: one-to-one, onto and bijective

The second concept of foundational importance that we shall look at in this chapter is that of a function.

Definition 2.3.1 (Function)**.** *A **function** f is a rule that assigns to each element of certain set A one* (and only one) *element from a set B. This is denoted by $f : A \to B$.*

The set A is called the **domain** of the function f and the set B is called the **codomain**. If a is an element of A then the element from B that f assigns to a is commonly denoted by $f(a)$ and is called the **image of a under f**. Also, given $b \in B$, an element $a \in A$ is called a **preimage of b under f** if $f(a) = b$. The **image** of f is the set $\{f(a) \in B \ : \ a \in A\} \subseteq B$.

A function $f : A \to B$ can also be expressed as a subset S of $A \times B$ such that if $(a, b) \in S$ and $(a, b') \in S$ then $b = b'$. This more abstract definition makes it clearer that the rule associating $a \in A$ to $b \in B$ can be completely arbitrary.

Example 2.3.2. When you count the elements of a finite collection (for instance, the students in your class) you are constructing a function from that finite collection to the set $\mathbb{N}$ of natural numbers.

Example 2.3.3. Let A be the set of all people in the world and let B be the set of all finite sequences of elements of $\{a, c, g, t\}$. Let f be the rule that assigns to each person in A their DNA. Then f is clearly a function from A to B, because each person has a unique DNA.

Example 2.3.4. Again, let A be the set of all people in the world, but this time, let B be the set of all their fingerprints. Then the rule that associates to each person in the world his/her fingerprints is a function $f : A \to B$.

Example 2.3.5. Let $\mathbb{R}$ (as usual) denote the set of all real numbers, let B be the set $[-1, +\infty)$,[2] and let f be the rule that to each real number x assigns its square x^2. Then f is a function.

[2]Mimicking the notation used for finite intervals, given a real number a, we shall denote by $(a, +\infty)$, $[a, +\infty)$, $(-\infty, a)$ and $(-\infty, a]$ the sets $\{x \in \mathbb{R} : x > a\}$, $\{x \in \mathbb{R} : x \geq a\}$, $\{x \in \mathbb{R} : x < a\}$ and $\{x \in \mathbb{R} : x \leq a\}$, respectively.

The image of 3 under f is 9. The number 7 has two pre-images: $\sqrt{7}$ and $-\sqrt{7}$. The number -5 has no pre-images. The image of f is $[0, \infty)$.

Definition 2.3.6. *A function $f : A \to B$ is said to be*

- ***one-to-one*** *if to different elements from A it assigns different elements from B. One-to-one functions are also called* ***injective****.*
- ***onto*** *if for every element in B there is an element in A whose image is precisely the given element from B.Onto functions are also called* ***surjective****.*
- ***bijective*** *if it is both one-to-one and onto.*

Example 2.3.7. The function described in Example 2.3.2 above is one-to-one because while counting the objects in a finite set you do not repeat any number. It cannot be onto because while counting the elements of a finite set you use only finitely many natural numbers.

Example 2.3.8. The function from Example 2.3.3 above is not onto because not every string of letters is a valid human DNA sequence. It is also not one-to-one because identical twins share the same DNA, so we will have in the set A different people with exactly the same image under f.

Example 2.3.9. The function from Example 2.3.4 above is a bijection. It is one-to-one because no two people share the same fingerprints (i.e., to different elements in A correspond different images). It is onto because each set of fingerprints belongs to somebody.

Example 2.3.10. The function $f : \mathbb{R} \to [-1, +\infty)$, defined by $f(x) = x^2$ $(x \in \mathbb{R})$. DRAW GRAPH. is not one-to-one because $-1 \neq 1$ but $f(-1) = (-1)^2 = 1 = 1^2 = f(1)$. It is not onto either, because -1 belongs to the codomain of f but there is no real number x such that $f(x) = -1$, i.e., -1 is not the image of any element under f.

Example 2.3.11. Let $f : [0, +\infty) \to [0, +\infty)$ be the function defined by $f(x) = \sqrt{x}$. DRAW GRAPH. Look at horizontal lines to count pre-images, check injective and surjective. It is onto because for every $y \in [0, +\infty)$ there is a number in $[0, +\infty)$, namely y^2, such that $f(y^2) = y$.

The function f is one-to-one because it is strictly increasing on $[0, +\infty)$, i.e., $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$.

ASIDE: To see this, note that $\sqrt{a} \leqslant \sqrt{b}$ implies that

$$a = \sqrt{a}\sqrt{a} \leqslant \sqrt{a}\sqrt{b} \leqslant \sqrt{b}\sqrt{b} = b.$$

So if $x_1 < x_2$, then $\sqrt{x_1} < \sqrt{x_2}$ (as otherwise we would have a contradiction).

Remark 2.3.12. To summarise a little:

- To show a map is one-to-one: you have to pick arbitrary $x, y \in A$ with $f(x) = f(y)$ and show that this implies $x = y$.

- To show a map is not one-to-one: you have to give specific $x \neq y$ in A with $f(x) = f(y)$.
- To show a map is onto: you have to pick arbitrary $b \in B$ and find an $a \in A$ with $f(a) = b$. There could be more than one a.
- To show a map is not onto: you have to pick specific $b \in B$ and show that there is no $a \in A$ with $f(a) = b$.

Definition 2.3.13. *Two functions f and g are said to be* ***equal****, denoted $f = g$, if they have the same domain and codomain, and $f(a) = g(a)$ for every element a in the domain.*

We need to be this precise about the domain, codomain and range, as we want to say that if f is one-to-one /onto and $f = g$ then g is one-to-one / onto.

Example 2.3.14. Consider the functions $f : \mathbb{R} \to [0, +\infty)$ defined by $f(x) = x^2 + 2x + 1$ $(x \in \mathbb{R})$, and $g : \mathbb{R} \to [0, +\infty)$ defined by $g(x) = (x+1)^2$ $(x \in \mathbb{R})$. They both have domain $\mathbb{R}$ and codomain $[0, +\infty)$. Moreover, for every x in the domain, $f(x) = x^2 + 2x + 1 = (x + 1)^2 = g(x)$. So $f = g$.

Example 2.3.15. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = |x| + 1$ $(x \in \mathbb{R})$, and let $g : \mathbb{R} \to [1, +\infty)$ be defined by $g(x) = |x|+1$ $(x \in \mathbb{R})$. Although with very similar appearances, the codomains of the two functions are different (the codomain of f is $\mathbb{R}$ while the codomain of g is $[1, +\infty)$), therefore, according to our definition, $f \neq g$.

In particular, g is onto and f is not.

A final comment on injectivity and surjectivity.

Lemma 2.3.16. *We can describe one-to-one and onto in terms of preimages.*

- *A function f is injective if and only if each element of b has at most one preimage.*
- *A function f is surjective if and only if each element of b has at least one preimage.*
- *A function f is bijective if and only if each element of b has exactly one preimage.*

We define next one of the simplest and most useful 'operations' one can perform with functions.

Definition 2.3.17 (Composition of functions)**.** *Let $f : A \to B$ and $g : B' \to C$ be real functions such that $B \subseteq B'$. The composition of f and g is the function $g \circ f : A \to C$, defined by the rule $(g \circ f)(x) := g(f(x))$ $(x \in A)$.*

Example 2.3.18. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = |x|$ $(x \in \mathbb{R})$, and let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = \sin(x)$ $(x \in \mathbb{R})$. Then $(f \circ g)(x) = f(g(x)) = f(\sin(x)) = |\sin(x)|$ $(x \in \mathbb{R})$. On the other hand, $(g \circ f)(x) = g(f(x)) = g(|x|) = \sin(|x|)$ $(x \in \mathbb{R})$.

Note from the last example that the order matters! In general, $f \circ g$ can be quite different from $g \circ f$. In fact, it may even be that one of the composite maps is defined and the other is not, as the following example shows.

Example 2.3.19. Let $f : (0, +\infty) \to \mathbb{R}$, $x \mapsto \ln(x)$, and let $g : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^3$. Then $g \circ f : (0, +\infty) \to \mathbb{R}$ is the map defined by $(g \circ f)(x) = g(f(x)) = g(\ln(x)) = (\ln(x))^3$ $(x \in (0, +\infty))$. However, $f \circ g$ is not defined, because the codomain of g (i.e., $\mathbb{R}$) is not contained in the domain of f (i.e., $(0, +\infty)$).

A simple but important fact about the composition of functions is its **associativity**. Precisely, if $f : A \to B$, $g : B' \to C$ and $h : C' \to D$ are such that $B \subseteq B'$ and $C \subseteq C'$ then $(h \circ g) \circ f = h \circ (g \circ f)$. Indeed, for every $a \in A$, one has that $((h \circ g) \circ f)(a) = (h \circ g)(f(a)) = h(g(f(a))) = h((g \circ f)(a)) = (h \circ (g \circ f))(a)$.

Definition 2.3.20 (Inverse of a function)**.** *Let $f : A \to B$ be a function. If there is a function $g : B \to A$ such that $(g \circ f)(a) = a$ for every $a \in A$ and $(f \circ g)(b) = b$ for every $b \in B$ then g is called an inverse for f and denoted f^{-1}.*

Example 2.3.21. Let $f : (0, +\infty) \to \mathbb{R}$ be defined by $f(x) = \ln(x)$ $(x \in (0, +\infty))$, and let $g : \mathbb{R} \to (0, +\infty)$ be defined by $g(x) = e^x$ $(x \in \mathbb{R})$. Then, for every $x \in (0, +\infty)$, $(g \circ f)(x) = g(f(x)) = g(\ln(x)) = e^{\ln(x)} = x$, and for every $x \in \mathbb{R}$, $(f \circ g)(x) = f(g(x)) = f(e^x) = \ln(e^x) = x$. So g is the inverse of f and f is the inverse of g.

Example 2.3.22. Let $f : [0, +\infty) \to [1, +\infty)$ be defined by $f(x) = (x + 1)^2$ $(x \in [0, +\infty))$, and let $g : [1, +\infty) \to [0, +\infty)$ be defined by $g(x) = \sqrt{x} - 1$ $(x \in [1, +\infty))$. Then $(g \circ f)(x) = g(f(x)) = g((x + 1)^2) = \sqrt{(x + 1)^2} - 1 = |x + 1| - 1 = (x + 1) - 1 = x$ $(x \in [0, +\infty))$, and $(f \circ g)(x) = f(g(x)) = f(\sqrt{x} - 1) = ((\sqrt{x} - 1) + 1)^2 = x$ $(x \in [1, +\infty))$. So g and f are each other inverses.

How can we decide whether a given function has an inverse or not?

Proposition 2.3.23. *A function $f : A \to B$ has an inverse if and only if it is bijective. Moreover, if f has an inverse then it is unique.*

Proof. Assume that $f : A \to B$ is bijective. We want to define a function $g : B \to A$. Let $b \in B$ and let a be the unique element of A with $f(a) = b$. We set $g(b) = a$. We must now check that g is an inverse to f. With b and a as above, it is clear that $f(g(b)) = f(a) = b$. Now pick $a \in A$, by definition $g(f(a)) = a$. Hence g is an inverse to f.

Now assume that $f : A \to B$ has an inverse $g : B \to A$. We must show that f is injective and surjective. Assume we have that $f(a) = f(a')$ for $a, a' \in A$. Then $a = g(f(a)) = g(f(a')) = a'$ and thus f is injective. Pick some $b \in B$. Then $f(g(b)) = b$, so f is surjective. Hence f is bijective.

Now we prove that any two inverses of a function $f : A \to B$ are equal. Assume $g, g' : B \to A$ are both inverses to f. Pick $b \in B$, then as f has an inverse it is bijective and there is a unique $a \in A$ with $f(a) = b$ Hence

$$g(b) = g(f(a)) = a = g'(f(a)) = g'(b)$$

and the functions g and g' are equal. □

Example 2.3.24. The function $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto \sin x$, has no inverse, because it is not one-to-one. For instance, $0 \neq \pi$ but $\sin(0) = \sin(\pi)$.

Example 2.3.25. The function $f : (0, +\infty) \to \mathbb{R}$, $x \mapsto x^2$, has no inverse, because it is not onto. Indeed, there is no number $x \in (0, +\infty)$ such that $x^2 = -1$. Note, though, that f is one-to-one, for if $x, y \in (0, +\infty)$ and $x \neq y$ then $x^2 \neq y^2$.

Example 2.3.26. Let $f : [0, +\infty) \to [0, +\infty)$, $x \mapsto \sqrt{x}$. We have seen above that this function is one-to-one and onto, and so, by Proposition 2.3.23, it must have an inverse. Its inverse is, as you probably suspect, the function $f^{-1} : [0, +\infty) \to [0, +\infty)$, $x \to x^2$. Indeed, we have that $(f \circ f^{-1})(x) = f(f^{-1}(x)) = f(x^2) = \sqrt{x^2} = |x| = x \ \ (x \in [0, +\infty))$ and $(f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(\sqrt{x}) = (\sqrt{x})^2 = x \ \ (x \in [0, +\infty))$.

2.4 Equivalence Relations

We want to give a mathematical structure for relations, the guiding examples are $\leqslant$, $\leq$, but other more exotic versions may be encountered (much) later on in your studies.

Definition 2.4.1. *A **relation** on a set A is a subset R of $A \times A$.*

If $(a, b) \in A$, we say that a and b are related, this is written in various ways such as $a \sim_R b$ or aRb.

Examples 2.4.2. Let $A = \mathbb{Z}$ and let R be the set of all pairs of integers (a, b) with $5 \mid (a-b)$. This defines arithmetic modulo 5, which you may have seen.

Let $A = \mathbb{R}$ and let R be the set of all pairs of real numbers (a, b) with $a < b$. This defines the relation 'less than'.

Let $A = \mathbb{C}$ and let R be the set of all pairs of complex numbers (a, b) with $|a| < |b|$.

There are many more we could include, and some look like a cheat, you have to know what $<$ means to define the set R for this. The point is that we have the formal definition in terms of subsets and products and the notation we actually use such as $<$, $\leqslant$, $\sim$ and so on. There are many kinds of special properties a relation can have.

Definition 2.4.3. *Let R be a relation on the set A, and a, b and c be elements of A.*

- *The relation R is said to be **reflexive** if for each $a \in A$, aRa.*
- *The relation R is said to be **symmetric** if whenever aRb, then bRa.*
- *The relation R is said to be **anti-symmetric** if aRb and bRa implies $a = b$.*
- *The relation R is said to be **transitive** whenever aRb, and bRc, then aRc.*

*The relation R is called an **equivalence relation** if it is reflexive, symmetric and transitive.*

Examples 2.4.4. Arithmetic modulo 5 on $\mathbb{Z}$ is an equivalence relation.

The relation $<$ on $\mathbb{R}$ is transitive and the relation $\leqslant$ on $\mathbb{R}$ is transitive and anti-symmetric.

Examples 2.4.5. The relation "has the same surname" is reflexive, symmetric and transitive.

The relation "is a sibling of" is symmetric and transitive. It isn't really reflexive as stated, but could be rephrased as has the same parents, which would be. But this is why words are tricky and lawyers are overpaid.

The relation "owes money to" is not reflexive. You can't really owe yourself money. It is not really symmetric or anti-symmetric. It is some sense transitive!

The relation "is a friend of" is not reflexive. People don't always like themselves. It is hopefully symmetric. It is not transitive, which is why sharing houses or parties can cause arguments.

The relation "is in love with" is also not reflexive. It is not symmetric, which is unrequited love, which is miserable. It is not transitive, which is a love triangle, which is even worse. In general if A loves B, and B loves C, then A usually really hates C.

Given a set A with an equivalence relation R, define a subset $[a] \subseteq A$ by $[a] = \{b \in A \ : \ bRa\}$. We call this the **equivalence class** of a. We can define a new set

$$A/R = \{[a] \ : \ a \in A\}$$

the set of equivalence classes of A (under R).

Examples 2.4.6. Arithmetic modulo $n \in \mathbb{N}$ on $\mathbb{Z}$ is an equivalence relation. The set of equivalence equivalence classes is denoted $\mathbb{Z}/n$.

$$A/R = \{[0], [1], \ldots, [n-1]\}$$

where $[a] = \{a + kn \ : \ k \in \mathbb{Z}\}$.

Now that we have covered the basics of logic, proofs, number systems and functions, we will move on to vectors. Much like the number systems we have studied, vectors have an addition, but they usually don't have an interesting multiplication. We will also study functions of vectors (known as linear transformations). Eventually we will be able to describe these linear transformations in terms of matrices. Matrices are somewhat like vectors and have both an addition operation and a (non-commutative) multiplication. This multiplication operation is not always invertible.

Matrices and vectors (and the real and complex numbers) are the basis of an area of mathematics called linear algebra, one of the fundamental subjects in mathematics. You will see more of this in your second year.

Let $f : A \to B$ be a function, let G be its graph: the subset of $A \times B$ of the form $(a, f(a))$. The map $A \to G$, $a \mapsto (a, f(a))$ is injective. The map $G \to B$, $(a, f(a)) \mapsto f(a)$ is injective if and only if f is. It is surjective if and only if f is.